



Nature-inspired computing and machine learning based classification approach for glaucoma in retinal fundus images

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Abstract

Glaucoma, commonly known as the silent thief of sight, is the second most common cause of blindness in humans, and the number of cases is steadily increasing. Conventional diagnostic methods utilized by ophthalmologists include the assessment of intraocular pressure using tonometry, pachymetry, etc. Yet, each of these evaluations is time-consuming, requires human involvement, and is prone to subjective errors. In order to overcome these hurdles, practitioners are studying retinal pictures for glaucoma diagnosis within the field of medical imaging. In addition, computer-assisted diagnosis (CAD) systems can be created to solve these obstacles by using machine learning approaches to classify retinal pictures as "healthy" or "infected." This work presents a reduced set of structural and nonstructural features(characteristics) to characterize pictures of the retinal fundus. The grey level co-occurrence matrix (GLCM), the grey level run length matrix (GLRM), the first order statistical matrix (FOS), the wavelet, and the structural features (like disc damage likelihood scale (DDLS) and cup to disc ratio (CDR)) are extracted. This set of features is sent to three classical soft computing algorithms (Particle Swarm Optimization (PSO), Artificial Bee Colony (ABC), and Binary Cuckoo Search (BCS)) and their two-layered model (PSO-ABC) to generate subset of reduced features (feature selection phase) that computes auspicious accuracy when sent to three machine learning classifiers (Random Forest (RF), Support Vector Machine (SVM), and Ensemble of RF, SVM, and Logistic Regression). According to our understanding, these four soft computing algorithms are rarely employed in this application field. For analyzing the performance of suggested strategy, the ORIGA, REFUGE, and their combinations are chosen as subject datasets. Standard statistical performance indicators, including accuracy, specificity, precision, and sensitivity, are calculated. The BCS delivers remarkable performance with a minimum of 91% accuracy and a maximum of 98.46% accuracy. PSO-ABC heavily decreases the original feature set, with minor accuracy sacrifices. The quantitative results are also compared in light of the most recent state-of-the-art published research. Owing to its exemplary performance, the suggested method will undoubtedly serve as a second opinion for ophthalmologists.

Keywords Glaucoma · Retinal fundus images · Structural features · Non-structural features · Particle swarm optimization (PSO) · Artificial bee colony optimization (ABC) · Binary cuckoo search (BCS)

1 Introduction

High intraocular pressure is the most commonly associated with the development of glaucoma, which affects the eye's optic nerve. Lucid fluid fills the front eye, known as "aqueous humor" fluid which is continually and regularly forms on the surface of the eyeball, providing a constant level of pressure [38]. The fluid does not drain from the eye because of the rise in pressure. Moreover the disease causes progressive impairment of the optic nerve responsible for sending image signals from the eye to the brain. Initially it has absolutely no effect on vision hence usually passes unnoticed by patients [90].

According to recent research, cataract is the first most frequent reason for poor vision followed by Glaucoma, according to the Glaucoma Foundation. Approximately, 3 million people in the Western Hemisphere have glaucoma but only a small percentage is aware of it [53]. Overall, in the world, there are more than 120,000 million cases of blindness, 9% to 12% of which are in the US [113]. African Americans are 15 times more likely to be visually impaired than Caucasians. In India, glaucoma patients are projected to be at 60 million. It is estimated that 12 million people are at risk of glaucoma and there are 1.2 million cases of blindness due to it each year, making the chances of getting glaucoma a major reason of permanent vision loss [112].

Normally, glaucoma screening is done by ophthalmologists using dimension of the fluid in the eye, the angle of drainage or out flow channel and visual analysis of retinal fundus image [33]. The analysis on these parameters is subjective in nature and may be human error-prone (intra-observer variability) depending upon the experience of the medical practitioner thus we require highly expertise doctors but there is a scarcity of it [55]. At the same time it is a time-consuming process which takes 20–45 min for diagnosis. Pre-screening for glaucoma using regular eye examinations thus becomes critical [54]. To decide whether glaucoma signs are present, an ophthalmologist will photograph the patient's retina. Given below, Fig. 1 presents the glaucoma infection scenario. Manual eye examinations can be very time-consuming. Additionally, photographs can be inspected by board-certified or specialist ophthalmologists to ensure accurate diagnosis [122]. There will be a lot of cases in which an ophthalmologist would like to look through an enormous numbers of images that aren't causing problems. Thus experts will be overburdened due to this exercise which requires a machine (like computer) based automated solution [34]. Automatic scanning systems are able to avoid these issues by sorting and finding the photographs of the glaucoma, alerting only those that are identified. Computing technology makes use of more advanced components, and recent image processing algorithms [94].

By comparison, however, manual activities and the assessment of any medical picture are tedious and open to interpretation. Therefore, there is a need for a computer aided diagnosis (CAD) which aids the ophthalmologists while they work on medical imagery [95]. Pre-processing and classification of medical images are required to take place before they are used for glaucoma diagnostic purposes. Toning down the input image for further analysis will alleviate problems with different filters and operations such as morphological are the first step to eliminate any forms of noise and defects from the picture. As an initial steps we cropped the images of the dataset, resized these cropped images in equal size of 224*224, convert these RGB images into Gray-scale and finally enhancing the quality of these images using CLAHE approach.

The classification is a decision-making approach to classify images as 'abnormal' or 'normal'. This work focuses on the machine learning approach to the classification of CAD structures. The CAD concept was introduced in 1966 and has been fully applied

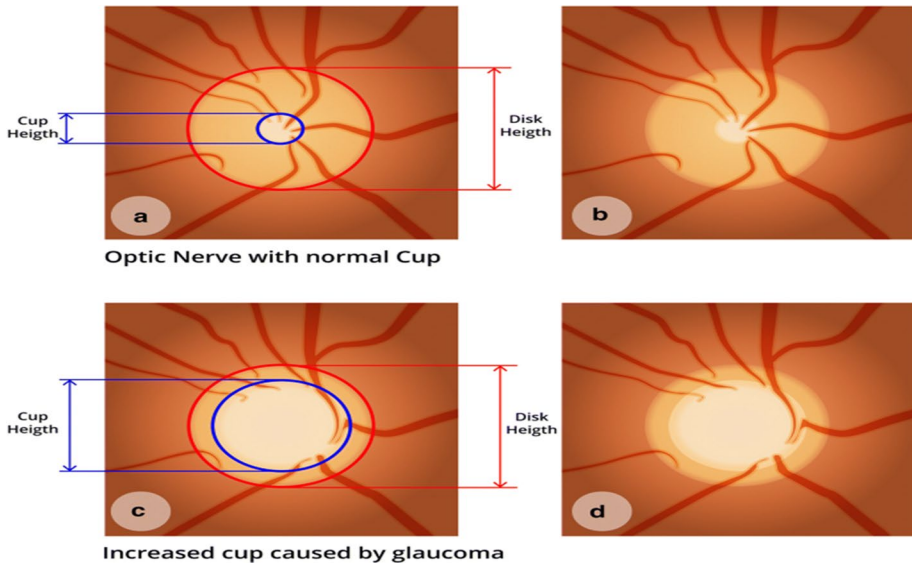


Fig. 1 Optic nerve with normal cup and increased cup caused by glaucoma: **a, b** optic nerve with normal cup and dimension quotes; **c, d** optic nerve with increased cup

since 1980 [29]. CAD systems have demonstrated their ability to support medical workers with high trust in making their decisions (such as triage, quantification and follow-up of promising cases). They also have advantages such as reproducibility and subtle improvements that are not apparent by visual examination. Glaucoma may be diagnosed by examination of fundus camera images. For example Fig. 1 presents the retinal image with absence of glaucoma (normal) and presence of glaucoma (abnormal). Here, it can be seen that an eye with glaucoma contains an enlarged area of optic cup occupying the optic disc. This work emphasizes on the classification module of CAD systems using blending of soft computing approaches with machine learning approaches in retinal images belonging to benchmark datasets. Feature selection methods are employed to shortlist significant features with high discriminatory power [72, 93]. The selection of robust features is necessary to smart model for any classification algorithm. In the design and development of machine learning algorithms, feature selection process is one of the basic step which aids in achieving improved classification accuracy in minimal computational time [107, 124]. Feature space with a large amount of irrelevant or redundant features may deteriorate the performance of the classifiers. Feature selection algorithms are employed to generate subset of most significant features from the original set to enhance the accuracy of the classifiers, by eliminating redundant features. In the prior works, researchers have proposed and implemented various bio-inspired computing approaches to lessen the dimensionality of the feature space. Studies indicate that the predictive ability of classification systems depends on the set of features [72, 93, 107, 121, 124]. The proposed methodology presents a PSO, ABC, BCS and 2-layered approach based optimization for feature selection in glaucoma infection. The effectiveness of the features selected is tested using 3 machine learning (RF, SVM and Ensemble) based algorithms as a classifier. In our empirical work these three optimization algorithms present multiple solutions fulfilling two criteria; achieving maximum

classification accuracy with a minimum number of features. Our efforts shortlist the vital and minimum number of features required for the confirmation of the disease.

The paper is organized as follows. Section 1 comprises of introduction to the disease along with discussion on Computer Aided Diagnosis system proposed in the study. Section 2 is focused on what previous work have been proposed by practitioners. Section 3 depicts the experimental set-up; dataset used; algorithms applied; focus on methodology (pre-processing, segmentation, feature extraction, feature selection and classification). Further, Section 4 explains the results and discussions comprising performance metrics, parameter settings of experimentation and performance investigation. Section 5 presents the comparison of this study with recent state-of-the-art approaches. At last, Section 6 concludes the study conducted.

The salient features of the proposed study are as follows

1. We have retrieved both structural and non-structural feature to construct initial features sets for disease classification from subject fundus images belonging to selected benchmark dataset(s). Our shortlisted soft-computing algorithms have rarely been applied earlier on these glaucoma datasets and on these features. Thus, our findings from this work are absolutely novel. Aside from the traditional implementations of ABC-Machine-Learning-Classifer (MLC), PSO-MLC, and BCS-MLC, we have also designed one new 2-layer approach (PSO-MLC-ABC-MLC) in all possible combinations of shortlisted MLCs to generate the reduced significant features set (out of the original 65 features) for glaucoma infection confirmation from fundus images of the benchmark dataset. These approaches are quick in nature, providing fast convergence with minimum computation, and are highly acceptable for their accuracy and glaucoma findings. As a result, the proposed work is used to evaluate a better classifier to classify glaucoma from a limited number of features.
2. The performance evaluation of BCS over two standard publically available datasets shows that it performs incredibly well in terms of efficiency for glaucoma diagnosis. Extensive experimentation has been performed (3 machine learning and 4 soft computing approaches have been employed on 3 combinations of 2 datasets) along with its thorough analysis. It is also observed that the BCS-PSO combination requires a few parameters for the implementation.
3. This study contributes to scientific research by recommending a novel classifier approach with incredible results: over 98% (up to 98.46%) accuracy in the detection of glaucoma disease for the binary class task. The classification accuracy of the proposed system is tested against two standard publically available and well-accepted datasets. This model would help ophthalmologists to diagnose and decide on the disease; it may also play the role of a second opinion. As one of the outcomes of this work, we are suggesting here a low-priced, quick, robust, easily implementable, and reliable intelligence system for screening this infection that can be employed for human society. As an impact, the vision loss rate may be reduced.
4. After reading Section 2 of this study titled "Related Work," it is clear that many studies have been published for glaucoma prediction and real-world problem solving using ABC, PSO, and cuckoo search. However, very few studies have been published in such a way that both structural and non-structural features are extracted, and this original feature set of 65 features has been reduced by these three algorithms to find (select) the most efficient features, which is the novelty of the work. This study also presents the results, interpretations, conclusions, and applicability in the most effective manner.

2 Related work

We have subdivided our extensively conducted survey into four subsections. Initial subsection is dedicated to some recent works on glaucoma infection identification from fundus images. Next three sub-sections are dedicated individually for PSO, ABC and Cuckoo applicability to solve various classes of problems.

Our in-depth survey has been broken into four subsections. Initial subsection is devoted to recent research on identifying glaucoma infections using fundus imaging. The following three subsections are dedicated separately to the applicability of PSO, ABC, and Cuckoo for solving diverse classes of problems.

For the classification of glaucoma utilizing various features retrieved from retinal pictures, a variety of methods have been utilized to date. A computational method for diagnosing glaucoma utilizing high order spectra (HOS) and textural properties was described by Acharya et al. [4]. Dua et al. [31] used SVM, SO, NB, and RF to recover wavelet features for glaucoma classification. Mookiah et al. [76] also utilized SVM to accurately identify glaucoma utilizing wavelet and HOS features. Noronha et al. [84] also obtained HOS and classified glaucoma using an SVM classifier and an NB classifier. Acharya et al. [5] identified glaucoma using Gabor features as well as SVM and NB classifiers. Issac et al. [49] also extracted structural data such as cup to disc ratio (CDR) and disc damage likelihood scale (DDLs) from a retinal image segmented for cup and disc. Later, Salem et al. [102] employed SVM for classification on the basis of extracted structural, textural, and intensity-based parameters for glaucoma diagnosis. Haleem et al. [46] employed features like structural, Gaussian, Gabor, and wavelets for distinguishing glaucoma. In addition, Claro et al. [21] gathered structural and textural properties; the maximum linear progression (MLP) classifier surpassed others classifiers in terms of accuracy. Singh et al. [114] also extracted wavelet features, after that, attributes (features) were chosen using principal component analysis (PCA) and evolutionary attribute selection (EAS). de Sousa et al. [26] extracted texture-based geostatistical functions such as semivariogram, correlogram, and covariogram. Koh et al. [62] extracted wavelet-based properties from retinal fundus images and employed random forest classifier. Septiarni et al. [106] collected statistical characteristics from a desired ROI in retinal images and categorized them using k-NN. In addition, Selvathi et al. [104] recovered wavelet-based characteristics for classification using NN. Shubhangi et al. [111] found cross entropy characteristics from wavelet-processed fundus image inputs and prioritized these traits using statistical testing. In addition, Renukalatha and Suresh [100] categorized glaucoma based on structural variables like CDR and non-structural aspects like texture.

Using an artificial neural network (ANN) classifier, Nayak et al. [79] diagnosed glaucoma by extracting CDR and the ratio of the area of blood vessels in the inferior-superior hemisphere to the quantity of blood vessels in the nasal-temporal hemisphere. According to Bock et al. [13], the images were initially preprocessed by changing the illumination and removing blood vessels, after that normalization was performed. Later, features were extracted using B-spline coefficients, Fourier coefficients, and raw intensity. The images were altered by Acharya et al. [5] using the Gabor transforms. Mean, variance, skewness, energy, kurtosis, Shannon, Renyi, and Kapur entropies retrieved from Gabor coefficients and SVM was used as a classifier. Raghavendra et al. [95] retrieved the segmented optic disc and applied the Radon transform to it. A modified census transform (MCT) was utilized to account for variations in light level. The MCT image was then exposed to the GIST descriptor to extract the spatial envelope energy spectrum and categorize it using the SVM

algorithm. Maheshwari et al. [69] accomplished image decomposition using 2D-variational mode decomposition (VMD). They applied Kapur entropy, Renyi entropy, and Yager entropy. The effectiveness of the automatically derived features from retinal fundus pictures for glaucoma diagnosis and detection was investigated [6]. The authors employed a representation or feature learning technique (Invariant Scattering Convolution Network or Wavelet Scattering Network) to the images. It is also investigated how the kind of 2-D channel picture and the wavelet scattering network parameter settings affect the detection accuracy. In order to segment and categorize retinal fundus pictures for the diagnosis of glaucoma, this research offers a computer approach [40]. The segmentation models are built on an attention U-Net with Inception-v3, VGG19, and Residual Neural Network 50 (ResNet50) CNN backbones. The three CNN architectures mentioned above are changed and used in the classification models as well. For precise pixel-by-pixel segmentation of the OC and OD, the authors of [44] suggested two networks: the separable linked segmentation network (SLS-Net) and the separable linked segmentation residual network (SLSR-Net). A large final feature map may be retained in SLS-Net and SLSR-Net, which improves the performance of OC and OD segmentation by reducing the loss of spatial information. For feature empowerment, SLSR-Net makes use of external residual connections. A separable convolutional connection was included in the suggested networks to increase computational effectiveness and lower network costs. Maheshwari et al. [68] used a 2D-empirical wavelet transform (2D-EWT) to transform the RGB image into sub-images. Correntropy is used as a feature, and LS-SVM was employed as a classifier. Karsu et al. [60] created a classifier employing CDR and anisotropic dual-tree complex wavelet transform features, in addition to a multi-layer perception (MLP) network. Sharma et al. [109] identified glaucoma with 14 statistical features taken from high order spectra and SVM as a classifier. Maheshwari et al. [70] collected local binary pattern-based statistical characteristics from each bit-plane of each RGB image channel, followed by statistical features derived from these characteristics and classification using an SVM classifier. In this study, Guru Prasad et al. [43], three kinds of segmentation techniques for the optic disc area of retinal fundus pictures are presented. For segmenting the optic disc, approaches based on thresholding, clustering, and regions were used.

Deep learning (DL) algorithms have recently been applied to a variety of biological imaging applications as a result of their powerful capacity for discriminative representation. Combining clinical measurement features with image-based features, a novel method for automatic glaucoma screening is proposed [42]. An upgraded UNet++ neural network is suggested to simultaneously segment the OD and OC depending on the region of interest (ROI). Optic cup to disc ratio are recovered from the segmentation outcomes. Then, the increasing field of view (IFOV) feature model is developed to fully extract image-based texture features, statistical features, and other hidden image-based characteristics. Training is then performed on a gradient boosting decision tree (GBDT) classifier for glaucoma screening. In this study, the authors [34] used the Bi-dimensional Empirical Mode Decomposition (BEMD) algorithm to breakdown the ROI into their constituent parts (BIMFs + residue). To extract features from decomposed BEMD components, VGG19 is implemented. Then, they combine the same ROI's features into a bag of features. Principal Component Analysis (PCA) is utilized to lessen the dimensions of characteristics. The acquired feature sets serve as input parameters for the SVM. In this paper [33], a CNN-based deep learning architecture was developed to classify fundus images. To extract discriminative characteristics from the fundus image, a CNN with 18 layers was developed and trained. There are four convolutional layers, two maximum pooling layers, and one fully linked layer. Using a segmentation-based methodology on the OD and OC,

this research [122] developed a unique method for identifying glaucoma. By tweaking the hyper-parameters and creating a custom loss function, a custom UNET++ model was created for the segmentation task. Using eight distinct ImageNet models, this experimental work [113] seeks to predict glaucoma from Optical Coherence Tomography(OCT). A comprehensive study is conducted to analyze the performance of these models on a variety of efficiency measures in order to determine the model with the highest efficiency. The four top-performing models were selected based on the results. Subsequently, these models were evaluated utilizing the well-known public standard Mendeley dataset. Using deep learning and a convolutional neural network, [38], the proposed framework determines the presence of glaucoma and analyses images of the retinal fundus. Utilizing the EfficientNet and UNet++ models, glaucoma was identified. Automatic glaucoma eye illness diagnosis was proposed in this study, [90], using deep learning (DL) and a deep-belief network (DBN). In addition, a contextualizing DL structure was proposed for obtaining multiple levels of fundus images to distinguish between glaucoma and non-glaucoma modes, with the network using the output of other CNNs as context information to support performance. The three existing machine learning models consisting of SVM, RF, and k-NN were executed and evaluated using tests.

This research, [55], utilizes machine learning/deep learning for the classification of glaucoma from fundus imaging and OCT. Using Major Voting and Weighted Decision Fusion, a unified framework has been proposed for this purpose, which combines predicted classes from machine learning and deep learning techniques. This study [54] introduces the classification of glaucoma network (CoG-NET), which is a deep network for predicting glaucoma. Moreover, corresponding feature activation maps for the proposed network validate its focus on the optical disc and cup exclusively present in the retinal fundus image for glaucoma diagnosis. The authors of [112] selected three public datasets, and examined their seven combinations. Initially, they developed five multimodal representations for each dataset. In the first vertical, authors extracted 36 crucial characteristics from each multimodal of a specific dataset. These extracted features are subsequently fused (known as early fusion) to generate the 180 features for each dataset. The top 50% of features are extracted to generate a feature vector. In the second vertical, they worked at the image level, sending images from each dataset's five multimodal dimensions to two deep learning classification techniques.

Based on this exhaustive assessment, it is clear that different researchers retrieved distinct classes of characteristics (structural or non-structural) for their experiments. Rarely have studies been conducted on the combination of structural and non-structural characteristics. Very few categorization researches used a soft computing-machine learning strategy. In addition, the use of classifiers and experimental datasets vary from study to study.

The next subsections discuss the application of PSO, ABC, and BCS to a variety of optimization issues. After reading this subsection, the reader will notice that PSO, ABC, BCS, and their variants have been successfully applied to other kinds of optimization problems, including the prediction of human diseases, but rarely to the optimization of features for glaucoma classification. In [105], the author utilized PSO and Association Rule Mining concepts to create a rule-based incremental categorization system. The incremental classifier was well-suited for sickness prediction on a rice disease dataset. This research [129] investigated three ensemble algorithms in conjunction with a feature subset selection (by PSO) to accurately predict the incidence of heart disease in a given patient. Based on actual coronary artery disease(CAD) data, the work [135] aims to offer a hybrid binary-real PSO that combines categorical and numerical particle encodings and a novel method for predicting particle velocity. This

paper [18] introduces a novel ensemble technique: Ensemble PSO (EnsPSO) which combines (i) Vote, (ii) Correlation-based Feature(s) Selection (CFS) technique, (iii) PSO algorithm, and (iv) random sampling method. This study [101] includes PSO as a strategy for feature selection in an effort to improve the breast cancer prediction accuracy of the model. This article [108] discussed a new model called as emotional neural networks (EmNNs) that is combined with the conventional PSO approach to detect coronary artery disease. The proposed model was also compared with adaptive neurofuzzy Inference system based on PSO (PSO-ANFIS). This article [115] provides an automatic categorization method for predicting leaf illness. Using PSO-based fuzzy c means segmentation (PSO-FCM) and Gaussian Mixture Model (GMM)-based background removal, the proposed technique segmented portions of diseased plant tissue. This article [127] proposed three new initialization strategies and three new personal best and global best updating mechanisms in PSO to develop novel feature selection approaches with the goals of maximizing classification performance, minimizing the number of features, and reducing computational time.

This article [48] describes a hybrid optimization method for the Feature Selection problem that combines the salp swarm algorithm (SSA) and PSO which yields the SSAPSO algorithm. This study's objective [134] was to introduce a modified PSO(AMPSO) technique for addressing multimodal multi-objective problems. This article [125] offers a hybrid PSO algorithm that combines an adaptive learning technique (ALPSO) with a competitive learning-based prediction strategy to accomplish algorithm exploitation. The practitioners [56] introduces psoCNN, a new method based on PSO for autonomously searching for suitable deep convolutional neural network (CNN) designs for image classification tasks that is capable of rapid convergence in comparison to existing evolutionary methods. This paper [133] provides a hybrid localization method based on particle filter and PSO, with a focus on issues requiring localization when an a priori environment map is available. This article describes a five-goal, multi-objective optimization model for the manufacture of green coal. The developed model utilized a hybrid multiple-objective particle swarm optimization (HMaPSO) method [23] to tackle the problem. Initially, the HMaPSO method was assessed on the DTLZ functions and its performance was compared to that of a number of well-known multi-objective algorithms. The HMaPSO technique was then utilized to solve the multi-objective optimization model for the production of green coal. This paper [10] suggests an upgraded variant of PSO known as restricted multi-swarm PSO without velocity constraints (CMPSOWV). Specifically, CMPSOWV includes a constraint management strategy to drive population searching toward viable portions of the search space before maximizing the objective function within these viable regions. The authors of [11] presented a hybrid approach (HFPSTO) that combines firefly and PSO. This study's objective [19] was to introduce a hybrid PSO with a spiral-shaped mechanism (HPSO-SSM) for selecting the optimal subset of features for classification using a wrapper-based technique. The objective of this paper [25] was to develop an ensemble model consisting of two pre-trained CNNs, namely VGG16 and VGG19, for classifying images of healthy and diseased leaves in order to diagnose plant illnesses. This research utilizes the orthogonal learning PSO(OLPSO) algorithm to optimize a number of hyperparameters by discovering their optimal values.

Notably, artificial intelligence has increased the significance of the ABC algorithm, and automated decision-making methods are now widely used. The purpose of this research [117] was to develop an automated diagnostic model for diabetes patients' diagnosis and prediction. The suggested diagnostic model, dubbed ABC-DNN-based diagnostic model, is constructed utilizing the ABC method and deep neural network (DNN) approach.

This research [27] examines the usefulness of the DNN method for resolving healthcare data imbalances. ABC-FS-optimized DNN was used to identify important characteristics of stroke disease. To obtain promising results in the categorization of medical data, a researcher [120] employs an orthogonal local preserving projection and an optimum classifier. The best results were generated through ABC and a neural network. Protein Structure Prediction (PSP) is a significant bioinformatics issue. Additionally, it has been shown that the Fitness Landscape Analysis (FLA) technique used in the ABC was appropriate for these situations. In light of these findings, the authors [103] of this article proposed new ABC variations called Chaotic Step Length-driven ABC algorithms dubbed Enhanced Chaotic ABC(ECABC). The position update phase of the onlooker bee phase was integrated with ten distinct Chaotic Step Lengths (CSLs) in these versions. Due to the critical significance of dimensionality reduction in identifying a plant leaf disease efficiently, this research [63] employs the singular value decomposition (SVD) method. The infected region was segmented using a fuzzy C means method based on artificial bee colonies. The purpose of this article [9] was to describe how to identify and classify illnesses in grape leaves using ABC-based feature selection. The optimum feature set was determined via ABC-based attribute selection.

In this study [59], ANFIS training uses an adaptive and hybrid artificial bee colony (aABC) algorithm. An adaptive neighborhood radius and arithmetic crossover were used to generate solutions in the aABC algorithm. This paper [47] proposed a new multi-objective ABC algorithm with non-dominated sorting and genetic operators. The proposed approach has two implementations: ABC with binary representation and ABC with continuous representation. To solve the well-known Quadratic Assignment Problem (QAP), researchers [30] offered hybrid ABC optimization methods. The suggested ABC method (ABC-QAP) and its parallel variant (PABC-QAP) were the first time the ABC meta-heuristic was used in conjunction with Tabu search to optimize the QAP. This article [45] presents a novel intrusion detection system (IDS) based on a multilayer perceptron (MLP) network, an artificial bee colony (ABC), and fuzzy clustering methods. The MLP identifies normal and abnormal network traffic packets, whereas the ABC method trains the MLP by maximizing the linking weights and biases. To aid in the optimum selection of Cluster Heads (CHs), authors present an Improved ABC Optimization-Based Clustering (IABCOCT) method in this article [35]. Practitioners [99] presented a new feature selection technique based on ABC and gradient boosting decision trees in this article, with the goal of resolving issues such as feature efficiency and informative quality. Using the bee colony technique to discover relevant characteristics, authors approach optimizes the decision tree's inputs globally. This study [96] examines how fractional derivatives and integration are used to solve an inventory management issue. Due of the model's complexity, numerical optimization using a modified ABC method was used instead of an analytical approach. This research [88] uses ABC to classify, improve, cluster, and segment medical pictures. The ABC technique for medical image analysis has good impacts on classification, segmentation, clustering, and enhancement procedures, and its usage has increased. This study may assist new researchers learn the ABC technique.

Due to its advantages, the Cuckoo Search Algorithm (CSA) has been successfully applied to a wide variety of optimization problems, including image processing [7, 12, 77, 89, 118], medical image processing [86]. Durgun and Yildiz [32] used the cuckoo algorithm to solve structural vehicle design optimization issues, resulting in increased fuel economy and reduced costs. Furthermore, Xiang-Tao and Ming-Hao [126] used CSA in conjunction with orthogonal learning to estimate chaotic system characteristics. Similarly, the binary version of CSA was used. To solve 0–1 knapsack problems, the same

binary versions were utilized [39]. The CSA binary version outperforms the binary PSO in terms of performance during evaluation. Additionally, CSA was utilized to optimize the design of truss structures and milling settings [130]. In mechanical engineering, CSA was effectively used to solve the problem of optimal synthesis of a six-bar doubled well linkage [14]. Additionally, CSA has been successfully adapted for structural engineering optimization issues, where the solutions found are significantly superior than the best solutions produced using existing techniques [119]. Yildiz [130] solved industrial optimization problems using a newly designed Cuckoo search method. As a result of CSA's applicability across fields, numerous hybridizations and modifications have been tried by various scholars [52, 57, 78] to address a variety of issues in important domains, such as energy conservation [91]. In the realm of power and energy, [2] utilized CSA to regulate the load frequency in a non-linear linked power system, resulting in improved settling times and different indices. The optimization of superconducting magnetic energy storage (SMES) system for automated generation management of a two-area thermal power system through CSA demonstrated the efficacy of fast-acting energy storage devices such as SMES [17]. Nogrehabadi et al. [83] used a hybrid Power series and Cuckoo Search Optimization technique (PS-CS) to solve a system of nonlinear differential equations originating from a distributed parameter model of a micro fixed-fixed switch subjected to electrostatic force and fringing field effect. Additionally, CSA adds to energy and cost reductions, distributed generation [64], plate fin heat exchangers [123], the short-term hydrothermal scheduling challenge [82], multi-reservoir system operation optimization [74], and economic load dispatch difficulties [81]. Authors of [98] presented an adaptive routing protocol based on CSA to offer safe and reliable routers between source and destination nodes, as well as to calculate the best distance and routing overhead. Authors of [80] utilized CSA to distribute a network reconfiguration with the purpose of reducing power loss and improving voltage profile. Furthermore, [16] utilized it to deduce the transition rules for cellular automata. Dhivya et al. [28] proposed a Cuckoo-based particle method for energy-efficient Wireless Sensor Networks with multimodal goal functions. The next work [20] investigated a novel meta-heuristic approach called Cuckoo Search Optimization (CSO) for constructing optimum and robust Power System Stabilizers (PSS) for multi-machine power systems (MMPS). Burnwal et al. [15] developed a cuckoo search-based strategy for scheduling flexible production systems with the objective of minimizing penalty costs associated with manufacturing delays and optimizing machine usage time. For binary optimization issues, Gherboudj et al. [39] presented a discrete binary cuckoo search (BCS) algorithm. There are subclasses of discrete optimization issues, such as discrete binary optimization problems, and their solutions are represented by a collection of bits, such as routing [131], job shop scheduling [92], and flow-shop scheduling problems [65]. Tein and Ramli [66] solved scheduling difficulties for nurses using the CS technique as the procedure of nurse scheduling is critical in healthcare facilities worldwide. Discrete CSA (DCSA) was offered as a solution for the classic Traveling Salesman Problem (TSP) in another paper [50]. Ouaarab et al. [86] presented how to solve the TSP using an enhanced and discrete version of the Cuckoo Search method. Ouyang et al. [87] introduced the DCSA algorithm for solving spherical TSPs with all points on the surface of a sphere. Ahmed T. Sadiq Al-Obaidi [8] offers a Cuckoo Search-based improved scatter search method. Scatter Search is used to do testing on the travelling salesman problem. This article [71] presents a novel method for addressing the graph colouring issue using the discrete Cuckoo Optimization Algorithm. A unique Cuckoo Search extension, snap-drift cuckoo search (SDCS), has been described that utilizes a learning method first and then considers enhanced search operators [97].

A modified cuckoo search method with rough sets was given in this article [1] for dealing with high-dimensional data via feature selection. To handle the features contained in the heart disease classification system, an artificial gravitational cuckoo search algorithm and a particle bee optimized associative memory neural network were implemented [51]. According to the behavior of the gravitational CS method, the dimension of the characteristics was reduced. The research analyses many characteristics of heart illness using the Artificial Gravitational Cuckoo Search Algorithm in conjunction with a Particle Bee Optimized Associative Memory Neural Network method. The authors of this article [3] provide a comprehensive method for diagnosing cardiac disease. Cuckoo search and rough set inference were combined in the proposed approach for inferencing decision rules. They have combined cuckoo search with the rough set (CSRS) for the purpose of knowledge inference. The suggested Rider-CSA method was aimed to identify plant diseases by combining the rider optimization algorithm (ROA) with Cuckoo Search (CS) [22]. The key geometric, texture, statistical, and intensity characteristics of the nodule of interest were retrieved in this study [93] from the precise segmentation of the Region of Interest (ROI). To categorize binary medical datasets, this article [75] proposed an improved cuckoo search-based extreme learning machine (ICSELM). In the realm of classification, the extreme learning machine (ELM) is a frequently used learning method for training single layer feed forward neural networks (SLFN). To increase the model's stability, an evolutionary algorithm enhanced cuckoo search was utilized to pre-train the ELM by choosing the input weights and hidden biases.

From this discussion and according to best of our knowledge (acquired from scholar.google.com) it can be concluded that very less (almost none) reports are published (available) on features optimization using PSO/ABC/BCS for glaucoma prediction; this investigation's primary aim is to fill this gap.

3 Proposed methodology

3.1 Materials and methods

ORIGA and REFUGE are two publicly available benchmark datasets containing the images used in the experiments. The total number of retinal fundus images in ORIGA is 650. The glaucoma images number 168 out of 650, while the healthy images number 482. Zhang et al. [132] utilize the publicly accessible dataset of retinal fundus images for glaucoma analysis for their experiments. The purpose of ORIGA is to assist the research community in optimizing techniques and creating new glaucoma analysis tools. REFUGE offers 1200 retinal fundus images for clinical challenges (Orlando et al., [85]). This competition consisted of three tasks: classification of glaucoma, segmentation of the optic disc and cup, and localization of the fovea.

The proposed methodology implements the methods for extracting a total of 65 features from the glaucoma fundus retinal images included in the aforementioned datasets. On the extracted 65 features, the PSO, ABC, and BCS optimization algorithms have been applied separately. The experiment was also conducted using a two-layered strategy consisting of PSO followed by ABC. This work presents a methodology with two primary phases. In the initial phase, feature extraction is carried out. Later, soft-computing algorithms are used as a feature selection algorithm and machine learning algorithms are used as a fitness evaluator. Images are categorized as either belonging to a healthy individual or a glaucoma

patient based on the features selected by soft computing algorithms. Tables 1 and 2 present the list and short definition about extracted features.

3.1.1 Structural features (cup diameter, disk diameter, cup to disk ratio, RIM, DDLS)

Prior to determining the cup-to-disc ratio, it is necessary to segment the optic disc and optic cup. In this procedure, the area of interest (ROI) is extracted from the input fundus image. The ROI is further subdivided into Red, Green, and Blue channels. The Red component is suitable for optic disc segmentation because of the abundance of information for disc area in the red channel, whereas the Green component is appropriate for optic cup segmentation due to the presence of the appropriate optic cup region in the green channel. Thus, the entire procedure is described with the assistance of Fig. 2, which itself is self-explanatory.

3.1.2 GLCM features

The gray-level co-occurrence matrix (GLCM), also known as the gray-level spatial dependency matrix, is a mathematical process of analyzing texture that recognizes the spatial association of pixels. The GLCM functions describe an image's texture by measuring how frequently pairs of pixels with unique values and in a given spatial relationship appear in an image, generating a GLCM, and extracting statistical measures from this matrix. The size of matrix $M_k \times M_k$, which describes the second-order joint likelihood of a mask-bound image field defined a $(x(l,n), \lambda, \theta)$. The (l,n) th element of matrix represents number of times the combined levels l and m occurs in two pixels within the image separated by distance λ along angle θ . Now, let $\mathfrak{F}$ be a small positive num $x(l,n)$ is the co-occurrence matrix for $\lambda=1$ and angle θ . $x_p(l) = \sum_{n=1}^{M_k} x(l,n)$ is marginal row probability. $x_q(n) = \sum_{l=1}^{M_k} x(l,n)$ is marginal row probability. η_p is the grey level intensity of x_p . η_q is the grey level intensity of x_q . σ_p and σ_q are the standard deviation of x_p and x_q . For each degree, the GLCM final value is calculated, and the mean of these values is returned. Horizontally the values of θ are 0° , the diagonally it represent 45° and 90° in vertical direction. Finally, the resulting matrix is calculated with the functionality described below. Table 3 presents the shortlisted GLCM features.

Table 1 List and category of extracted features

S. No	Type	Name
1	Structural Features	CDR,DDLS,RIM Width,Cup Diameter,Disk Diameter(5)
2	GLRM	SREE,LRRE,GLNN,RPP,RLNN(5)
3	GLCM	cntra, corrlp, autocorr, cpromm, cshad, dissii, maxpr, sossvhh, energ, entroo, homopp, savvghh, svarh, denthh, senhh, dvaarhh, inf22hh, indncc, idmncc(19)
4	FOS	Entropy, Mean,Standard Deviation,Variance,Kurtos,Skewness(6)
5	Wavelet	WF_f1,WF_f2, WF_f3,WF_f4, WF_f5,WF_f6, WF_f7,WF_f8, WF_f9,WF_f10, WF_f11,WF_f12, WF_f13,WF_f14, WF_f15,WF_f16, WF_f17,WF_f18, WF_f19,WF_f20, WF_f21,WF_f22, WF_f23,WF_f24, WF_f25,WF_f26, WF_f27,WF_f28, WF_f29,WF_f30(30)

Table 2 List and short definition of the features

S. No	Type	Description
1	CDR	It is a Cup-to-Disc ratio in fundus images
2	DDL5	It is Disc Damage Likelihood Scale. The DDL5 is a method for determining the degree to which glaucoma has contributed to the degeneration of the optic nerve
3	RIM width	Between the optic disc and the optic cup lies a region known as the retinal rim. This is the border of the retina
4	Cup Diameter	It is the total no. of pixel horizontally or vertically in Optic Cup region
5	Disc Diameter	It is the total no. of pixel horizontally or vertically in Optic Disc region
6	Autocorrelation	Autocorrelation is a valuable method in signal processing for detecting recurring patterns in a signal. This can be accomplished in a variety of ways
7	Contrast	The color or grayscale disparity between numerous visual characteristics in both analog and digital images
8	Correlation	The difference in color or grayscale between several visual features in both analog and digital photographs
9	Cluster prominence	Cluster prominence is another indicator of asymmetry. When the cluster prominence value is high, the image becomes less symmetric. Furthermore, the GLCM matrix exhibits a peak around the mean values when the cluster prominence value is low
10	Cluster Shade	Cluster shading, which is a measurement of the skewness of the matrix, is also presumed to investigate the perceptual concepts of homogeneity
11	Dissimilarity	The degree to which two things (pixels) in the region of interest differ from one another can serve as a measurement for the distance that separates them from one another.
12	Energy	The result produced by the GLCM's 'Energy' function is the sum of squared elements
13	Entropy	Entropy is a measurement of how random the intensity levels of a neighborhood are
14	Homogeneity	After calculating a value that quantifies how close the distribution of items in the GLCM is to the diagonal of the GLCM, it then returns that number. The value that it calculates can be thought of as a distance measure
15	Maximum Probability	It measure the maximum value in GLCM matrix.
16	Sum of squares variance	The squared sum of the elements that make up the GLCM is what this function will return.
17	Variance	It gives a depiction of the squares of the standard deviations that correspond to the values of the output and input images respectively. Depending on the format, an image that has a variance can have a number of different variations that are easily distinguishable from one another.

Table 2 (continued)

S. No	Type	Description
18	Entropy	The discrete entropy refers to the number of bits that are necessary for the encoding of visual data. This quantity can vary depending on the type of data being encoded.
19	Short run emphasis	This metric rises when short runs predominate, as is the case, for instance, in fine-grained textures
20	Long run emphasis	This measure goes up when long runs are predominant, which occurs, for instance, in coarse textures or textures with broad swaths of material that are all the same
21	Grey level run emphasis	Having a gray level run refers to a line of pixels that are all moving in the same direction and all have the same value for their intensity. This type of run can only occur in particular directions. A run that falls within this category is called a gray level run

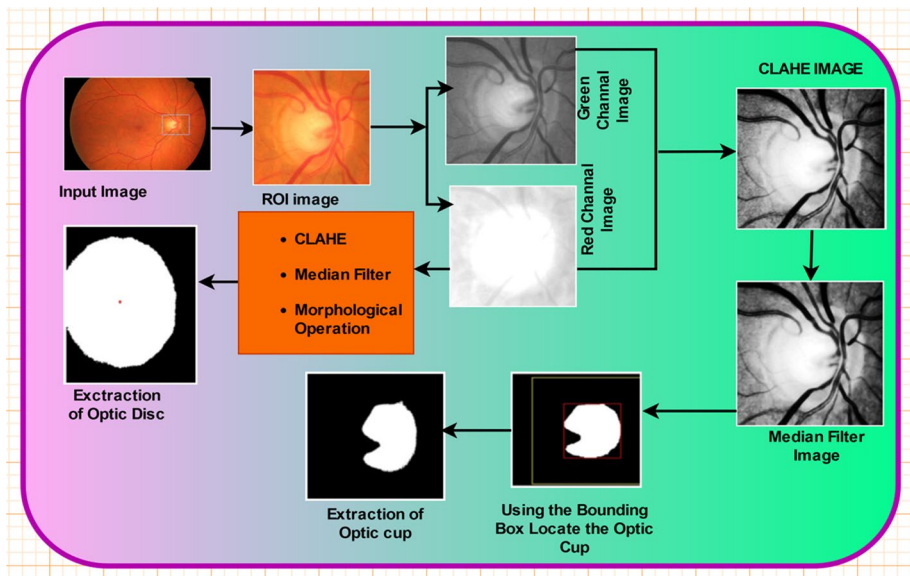


Fig. 2 Sample for extraction optic cup and optic disc from Retinal fundus image

3.1.3 Grey level run length matrix (GLRM)

The GLRM is a matrix that provides details regarding textural features for analyzing an object's texture. It's known as a line of pixels in a specific direction with the same intensity values. The sum of such pixels is referred to as the grey level run length, and the number of times it occurs is referred to as the run length value. In GLRM, $x(l, n|\theta)$ the (l, n) th element defines the number of runs with grey level l and n in the image along θ . Now let M_k be the discrete value of intensity in the image. M_r be the discrete run lengths in the image. M_p be the voxels number in the image. $M_r(\theta)$ is the number of runs along angle θ . Table 4 presents the shortlisted GLRM features.

3.1.4 First order statistical (FoS) features

FoS features can often used to evaluate an image's texture by the computation of an image histogram showing the likelihood of a pixel appearing in an image. These characteristics rely only on individual pixel values and not on the relationship with the pixels nearby. The first order histogram estimate is given by $p(c) = \frac{N(c)}{K}$. Here c represents the grey level in the image, K is the total number of pixels in the neighborhood window centered around the expected pixel and $N(c)$ is the number of pixels of gray value c in the window $0 \leq c \leq M - 1$.

The FoS characteristics are widely used as described below: mean, standard variation, variance, curtosis, skewness and entropy. Table 5 presents the shortlisted FoS features.

Table 3 Shortlisted GLCM features

S.No	Features	Equation
1	Autocorrelation	$Autocorr = \sum_{l=1}^{M_k} \sum_n^{M_k} x(l, n) \ln$
2	Contrast	$cntra = \sum_{l=1}^{M_k} \sum_n^{M_k} (i - j)^2 x(l, n)$
3,	Correlation	$corrIp = \frac{\sum_{l=1}^{M_k} \sum_{n=1}^{M_k} x(l, n) \ln - \eta_p \eta_q}{\sigma_p(l) \sigma_q(n)}$
4	Cluster prominence	$cpromm = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} (l + n - \eta_p - \eta_q)^4 x(l, n)$
5	Cluster Shade	$cshadd = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} (l + n - \eta_p - \eta_q) 3x(l, n)$
6	Dissimilarity	$dissii = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} l - n x(l, n)$
7	energy	$energ = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} x(l, n)^2$
8	Entropy	$entroo = - \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} x(l, n) \log(x(l, n) + \mathfrak{I})$
9	Homogeneity	$homopp = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} \frac{x(l, n)}{1 + l - n ^2}$
10	Maximum Probability	$\maxpr = \max(x(l, n))$
11	Sum of squares variance	$sossvhh = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} (l - \eta_p)^2 x(l, n)$
12	Sum variance	$smmv = \sum_{r=2}^{2M_k} (r - SA)^2 x_{p+q}(r)$
13	Sum entropy	$senthh = \sum_{r=2}^{2M_k} x_{l+n}(r) \log(x_{p+q}(r) + \mathfrak{I})$
14	Difference variance	$dvaarhh = \sum_{r=2}^{M_k-1} (r - DA)^2 x_{p-q}(r)$
15	Difference entropy	$denthh = \sum_{r=2}^{M_k-1} x_{p-q}(r) \log(x_{p-q}(r) + \mathfrak{I})$
16	Information measure of correlation2	$\inf \ 22hh = (1 - \exp[-2(HXY2 - HXY)])^{\frac{1}{2}}$
18	Inverse Difference normalized	$indncc = \sum_{r=0}^{M_k-1} \frac{x_{p-q}(r)}{1 + (\frac{r}{M_k})}$
19	Inverse difference moment normalized	$idmncc = \sum_{l=1}^{M_k} \sum_{n=1}^{M_k} \frac{x(l, n)}{1 + l - k ^2}$

Table 4 Shortlisted GLRM features

S.No	Features	Equation
1	Short run emphasis	$SREPP = \frac{\sum_{l=1}^{M_k} \sum_{n=1}^{M_k} \frac{x(l,n \theta)}{n^2}}{M_r(\theta)}$
2	Long run emphasis	$LREPP = \frac{\sum_{l=1}^{M_k} \sum_{n=1}^{M_k} x(l,n \theta)n^2}{M_r(\theta)}$
3	Grey level run emphasis	$GLNN = \frac{\sum_{l=1}^{M_k} \sum_{n=1}^{M_k} (x(l,n \theta))^2}{M_r(\theta)}$
4	Run percentage	$RPP = \frac{M_r(\theta)}{M_p}$
5	Run length non-uniformity	$RLNU = \frac{\sum_{n=1}^{M_k} \sum_{l=1}^{M_k} x(l,n \theta)}{M_k(\theta)}$

3.1.5 Discrete wavelet transforms (DWT) features

The transformation of wavelets is the division of data, operators or functions into different frequency components and the analysis of each with a resolution corresponding to a scale. Here, Wavelet transforms the picture into four distinct elements, i.e. approximation, horizontal, diagonal and vertical. Approximation variable is used for scaling and three for localization. For images, separate wavelet Transform with One-Dimension filter bank is added to rows and columns of each channel. Here for p rows and q columns, we have suppose the scaling function $\mathfrak{F}_{m,p,q}(r,s)$ and for translation function $\mathcal{G}_{m,p,q}^l(r,s)$ in Equation () and (). The Horizontal high-pass channels, vertical high-pass channels and diagonals high pass channels are $\zeta^H(r,s), \zeta^V(r,s), \zeta^D(r,s)$. All these important channels are extracted from $\mathfrak{F}_{m,p,q}(r,s)$ scaling function.

$$\mathfrak{F}_{m,p,q}(r,s) = 2^{m/2} \mathfrak{F}(2^m r - p, 2^m s - q) \quad (1)$$

Table 5 Shortlisted FoS features

S.No	Features	Equation
1	Mean	$\bar{c} = \sum_{c=0}^{M-1} c.p(c)$
2	Standard deviation	$\sigma_c = \sqrt{\sum_{c=0}^{M-1} (c - \bar{c})^2 c(p)}$
3	Variance	$\sigma^2 = \sum_{c=0}^{M-1} (c - \bar{c})^2 c(p)$
4	Kurtosis	$krr = \frac{1}{\sigma^4} \sum_{c=0}^{M-1} (c - \bar{c})^4 p(c) - 3$
5	Skewness	$skk = \frac{1}{\sigma^3} \sum_{c=0}^{M-1} (c - \bar{c})^3 p(c)$
6	Entropy	$entr = \sum_{c=0}^{M-1} p(c) \log p(c)$

$$\phi_{m,p,q}^l(r, s) = 2^{m/2} \phi(2^m r - p, 2^m s - q) l = \{H, V, D\} \quad (2)$$

Here the three main components of wavelets are diagonal (D), horizontal (H) and vertical (V). For the image $f(r, s)$ of the size p, q , The discrete wavelet transform is as follows:

$$W_{\mathfrak{S}}(j_0, p, q) = \frac{1}{\sqrt{pq}} \sum_{r=0}^{p-1} \sum_{s=0}^{q-1} f(r, s) \mathfrak{S}_{j_0,p,q}(r, s) \quad (3)$$

$$W_{\mathfrak{S}}(j_0, p, q) = \frac{1}{\sqrt{pq}} \sum_{r=0}^{p-1} \sum_{s=0}^{q-1} f(r, s) \phi_{m,p,q}^l(r, s), l = \{H, V, D\} \quad (4)$$

j_0 is an arbitrary starting scale and $W_{\mathfrak{S}}(j_0, p, q)$ coefficient define an approximation of $f(r, s)$ at scale j_0 . The $W_{\phi}^l(m, p, q)$ coefficients add horizontal, vertical and diagonal details for scales $m \geq j_0$. We normally let $j_0 = 0$ and select $p = q = 2^m$ so that $m = 0, 1, 2, \dots, m-1$.

3.2 Algorithms and their implementations

3.2.1 PSO

As a heuristic approach to global optimization, the concept of particle swarm optimization was first introduced by Doctor Kennedy and Eberhart in the year 1995 [61]. Swarm intelligence is the source of this concept, which was developed from research on the behaviour of flocking populations of fish and birds. Before settling in one location, the birds will either disperse in all directions or gather in large groups to search for food. One of the birds can always smell the food extremely clearly, despite the fact that the other birds are moving from one location to another in search of food. This suggests that the bird is aware of where the food can be found, exhibiting greater knowledge of the available food resources. The birds will eventually congregate at the spot where food can be acquired because they are always passing information, particularly excellent information, from one location to another in their search for food. This is because the birds are constantly exchanging information. In terms of the particle swarm optimization algorithm, the development of the solution swarm is analogous to the migration of birds from one location to another, high-quality information is equivalent to the most optimal solution, and food resources are analogous to the most optimal solution throughout the entirety of the course.

In the most fundamental version of the particle swarm optimization method, the particle swarm is made up of "n" individual particles. The location of each individual particle in D-dimensional space suggests a potential solution to the problem that is being solved by the algorithm. Each particle's position within the swarm is affected by the most optimistic particle's position during movement (individual experience) and position in its surroundings (near experience). When a swarm encircles a particle, the swarm's most hopeful location also happens to be where the particle is. The name of this algorithm is the complete PSO. The partial PSO approach is the name for the method that makes use of a small area [110]. Each particle tries to modify its position by using the current position, current velocities, the distance between the current position and personal best.

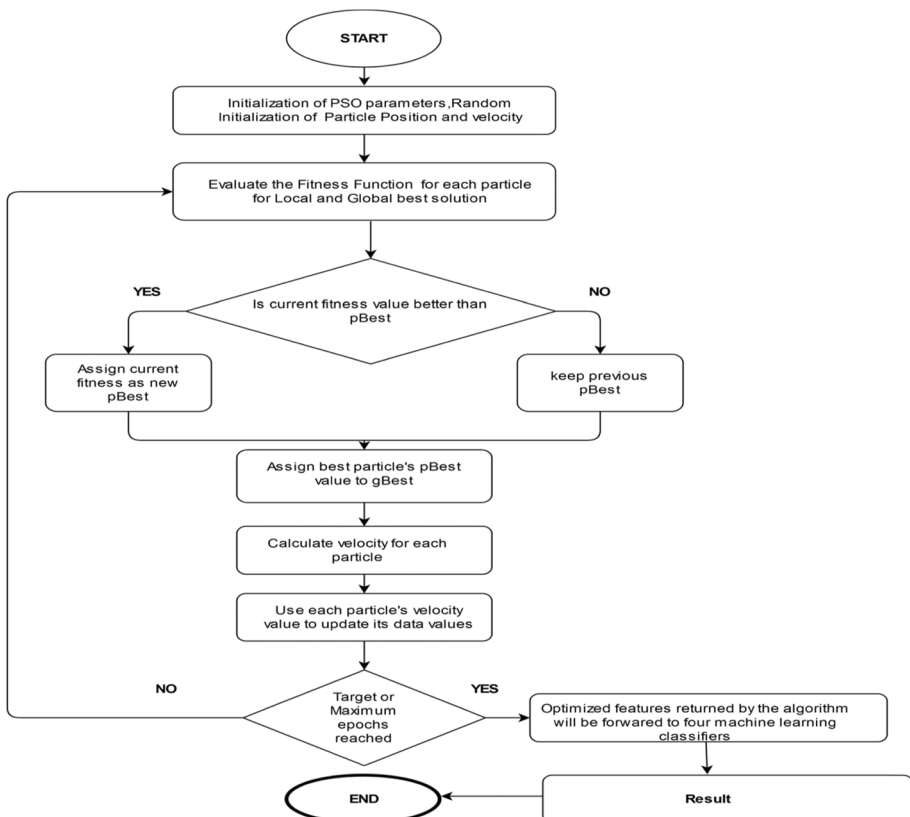
Table 6 Parameters used in PSO

dimension=65	No_of_iterations = 10,000
number of particles in the swarm = 50	'c1': 2 (cognitive parameter)
'c2': 2 (social parameter),	'k': 5 (number of neighbors to be considered. Must be a positive integer less than:code:`n_particles`)
'p':2 (the Minkowski p-norm to use. Euclidean distance)	'c2': 2 (social parameter)

$$v_{id1}^{k1+1} = wv_{id1}^{k1} + c_1r_1^{k1}(pbest_{id1}^{k1} - x_{id1}^{k1}) + c_2r_2^{k1}(gbest_{d1}^{k1} - x_{id1}^{k1}) \quad (5)$$

In this equality v_{id1}^{k1+1} is representing the current velocity, v_{id1}^{k1} representing the previous velocity here $id1$ representing the swarm like $id1 = 1, 2, 3, \dots, N$. It is the index of each particle. $k1$ is the iteration number. The values of $c1$ and $c2$ lies between 0 and 1.

New position is updated by the following equation:

**Fig. 3** Flowchart of implemented PSO

$$x_{id1}^{k1+1} = x_{id1}^{k1} + v_{id1}^{k1+1} \quad (6)$$

Table 6 shows the parameters that have been initialized in PSO (Fig. 3).

Objective function:

$$F_n = [\text{Max}[\beta \times \text{Max}(\text{ACC}) + (1 - \beta) \times \text{Min}(\frac{\text{Number of selected features}}{\text{Total number of features}})]$$

3.2.2 ABC

ABC is developed based on inspecting the behaviors of real bees on finding nectar and sharing the information of food sources to the bees in the hive [58].

In the ABC algorithm, the colony of artificial bees contains three groups of bees:

- The Employed Bee:

It stays on a food source and provides the neighborhood of the source in its memory.

- The Onlooker Bee:

It gets the information of food sources from the employed bees in the hive and select one of the food source to gathers the nectar.

- The Scout:

It is responsible for finding new food, the new nectar, sources.

3.2.3 Distribution of bees

In the ABC algorithm, first half of the colony consists of employed artificial bees and the second half constitutes the onlookers. For every food source, there is only one employed bee. In other words, the number of employed bees is equal to the number of food sources around the hive. The employed bee whose food source is exhausted by the employed and onlooker bees becomes a scout (Fig. 4).

3.2.4 Cuckoo search algorithm

The Cuckoo search algorithm is a meta-heuristic algorithm that takes its cue from some cuckoo species that lay their eggs in the nests of other species of birds [128]. They lay their eggs in the nest of other species, which boosts their chances of surviving and producing more offspring. The two bird species in this method are cuckoo birds and host birds (Other Species) (Fig. 5). It is possible for these birds to lay their eggs in host nests, and they are even able to mimic the outer characteristics of host eggs, such as their color and spotting. If the host tries this method and it doesn't work, they can either throw away the cuckoo's egg or give up and start a new nest somewhere else. Basically for implementing the cuckoo search three rules has been defined. The first one is each Cuckoo Lays only one egg at a time and places it in a randomly

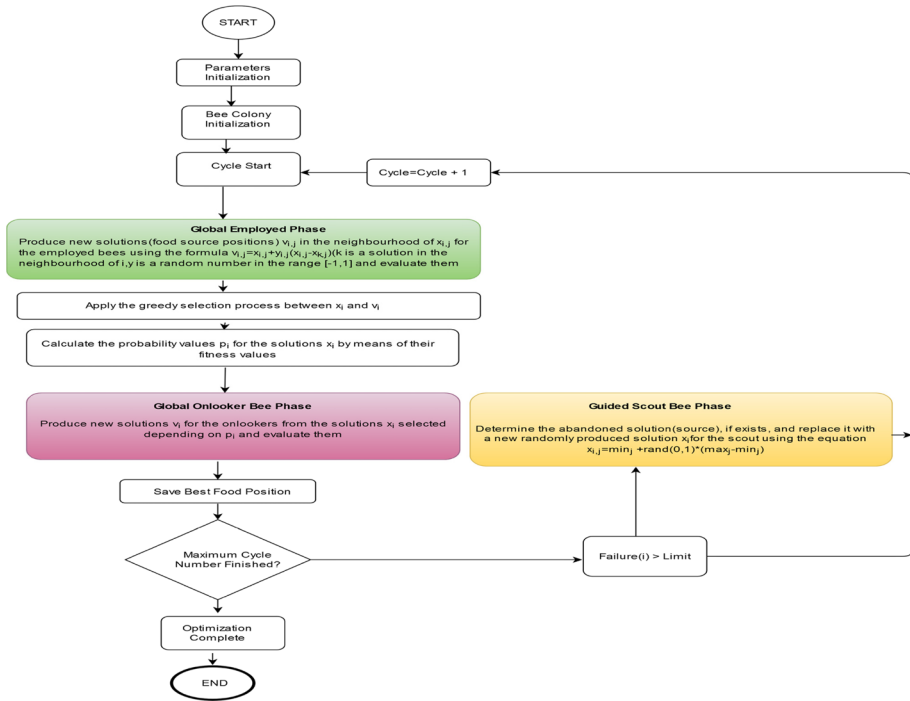


Fig. 4 Flowchart of implemented ABC

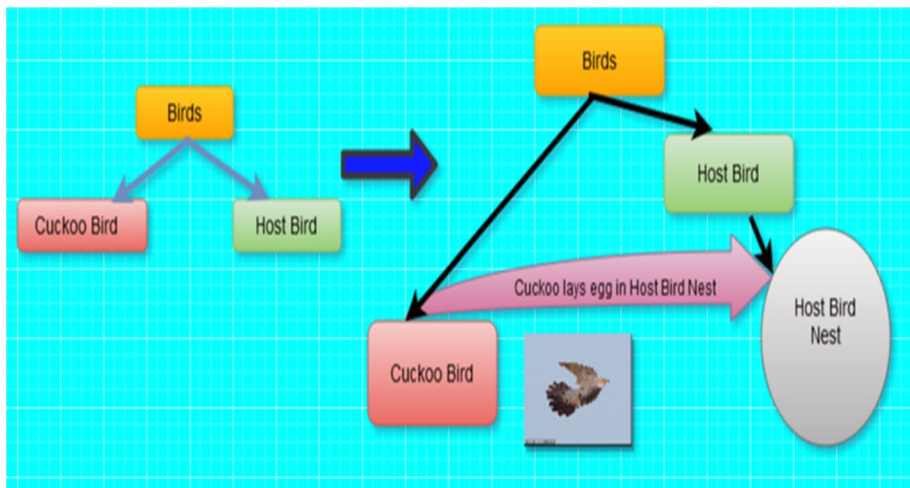


Fig. 5 Basic functioning of cuckoo birds

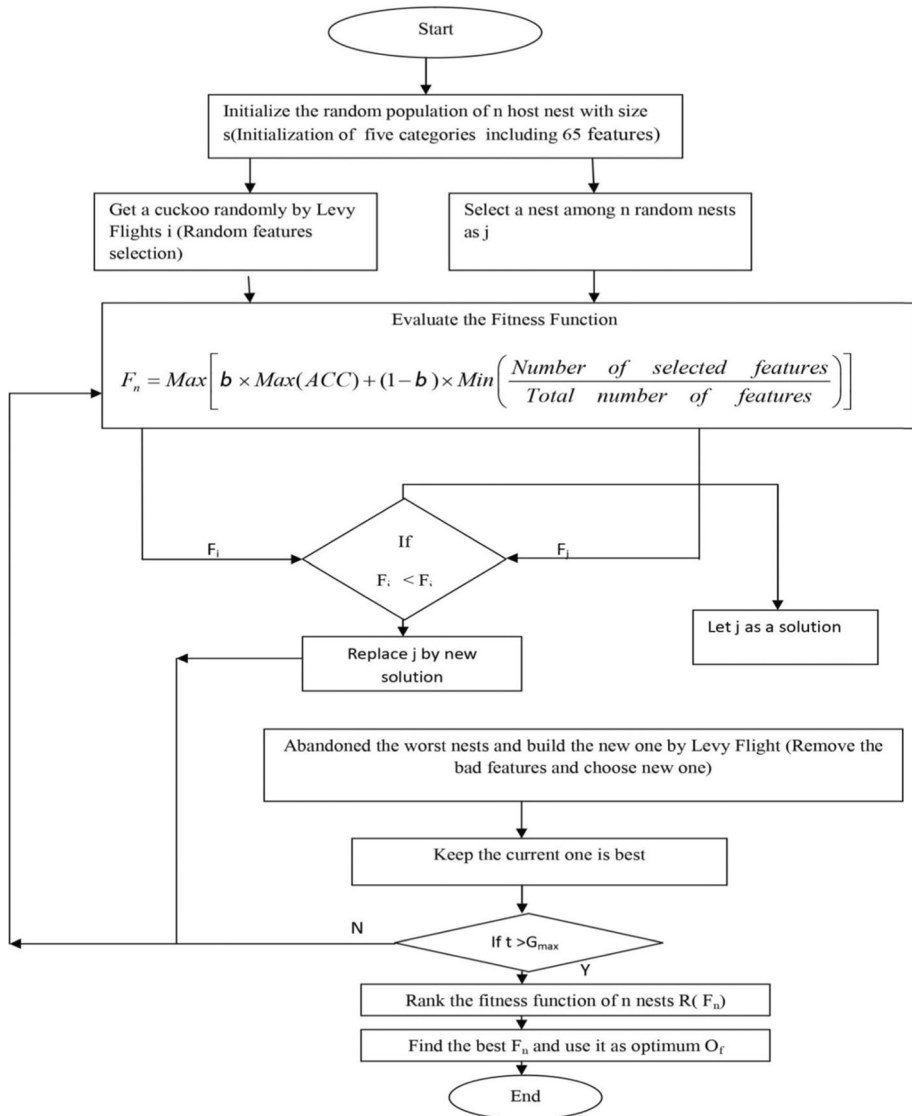


Fig. 6 Flowchart of implemented Binary Cuckoo Search (BCS) algorithm

selected nest. The Second one is the Best Nest with high quality of Eggs will carry over to next generation and the third rules is the number of available host nest is fixed. Host Bird discovers cuckoo eggs with probability $P_a \in (0, 1)$. Host bird can throw away the egg or leave nest/build new one. The parameters used in the cuckoo search optimization algorithm. The parameters of CS optimization algorithm are given as, number of nest $n = 50$, nest size = 65, minimum number of generation $t = 0$, $P_a = 0.25$, step size $\alpha = 0.01$, $\lambda = 1.5$, $\beta = 0.75$, Maximum generations $G_{\max} = 10,000$. The Fig. 6 shows the implementations detail of cuckoo search.

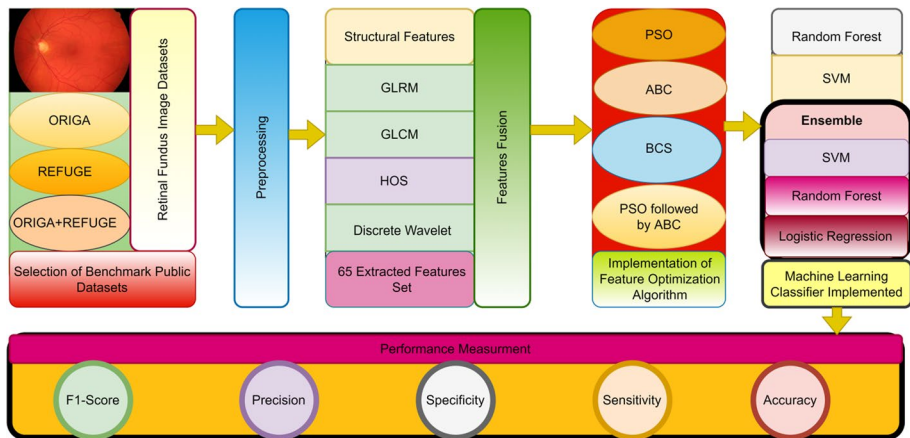


Fig. 7 Diagrammatic representation of the whole implemented work

3.2.5 2-layered optimization approach

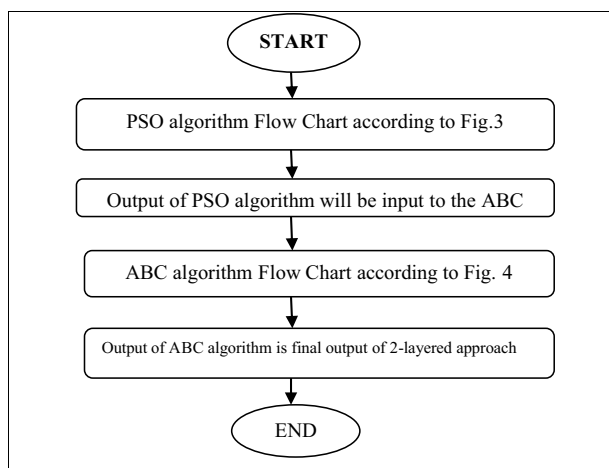
We have also implemented a 2-layered approach consisting of the PSO algorithm followed by the ABC algorithm (Fig. 7). The output of PSO (features returned by the PSO) will be input to the ABC. Figure 8 represents the flow chart of the working of this approach. Meanwhile, the whole layout of this presented work is shown with the help of Fig. 7.

4 Results and discussion

In this section detailed results are presented in tabular format as well as graphically. The detailed discussion on the results is also presented side-by-side.

First, it's important to know the following concepts to understand how useful clinical tests are: In clinical testing, the terms specificity and sensitivity are used. The population

Fig. 8 Flowchart of implemented 2-layered PSO-ABC algorithm



under test has no impact on them. The positive and negative predictive values of a test are what matter to a doctor when evaluating its value. They are based on the target population's propensity for sickness. In order to make a diagnosis, many clinical tests are carried out to confirm or rule out the existence of an illness. To gauge a test's clinical relevance, positive and negative predictive values are important. Both those who are well and those who are ill should be accurately identified.

True positive: Both patient and test are positive. The program successfully recognized these data points as positive.

False Positive: Patient has no illness, yet the test is positive. Incorrectly recognized data points are those whose real results were bad.

True negative: Patient is free of illness and test results are negative. The program accurately recognized these data points as negative.

False negative: Patient has illness but test comes back negative. Incorrectly recognized data points are those whose real results were positive.

Sensitivity analysis is used in studies to determine how reliable clinical trial results are. They support the interpretation or validation of findings. This is particularly true if the findings are robust to a variety of presumptions, methods, and circumstances. Sensitivity studies are a major component of studies assessing the reliability of primary data analysis in clinical trials. A test's sensitivity is how well it can identify sick people. They support the interpretation or validation of findings. This is particularly true if the findings are robust to a variety of presumptions, methods, and circumstances. Sensitivity analysis plays a significant role in studies assessing the reliability of primary data analysis in clinical trials. The capacity of a test to reliably identify sick people is known as sensitivity. The number of people who test positive for a condition determines how sensitive a medical test is to detecting it. Each and every person who tests positive for the disease will be found by a test with 100% sensitivity. Clinical tests that are highly sophisticated may be able to accurately identify sick people. Every case is correctly diagnosed by a test with 100% sensitivity. In 80% of tests with 80% sensitivity, true positives are detected (false negatives). Extreme sensitivity is necessary when a test is used to identify a serious but treatable illness (e.g., cervical cancer). A test has a 100% chance of correctly identifying every person who has a disease, according to the true positive rate (TPR).

The ability of a test to correctly rule out healthy patients is referred to as specificity. The ability of a clinical test to accurately identify healthy people is thus related to its specificity. The number of people who test negative but don't have the disease is referred to as a test's specificity. A 100% specificity test will find all of them if a patient tests negative. The ability to recognize healthy people is what determines how specific a clinical test is. As a result, a test with 100% specificity can identify every healthy person. Nevertheless, a test with 80% specificity misclassifies 20% of healthy people as positive (false positives). Based on the cut-off value, a quantitative test's sensitivity and specificity are determined.

The accuracy of a test in statistical binary classification analysis is measured by the F1 score, also known as the F-score or F-measure. It evaluates categorization schemes that assign occurrences a "positive" or "negative" label. It is calculated by dividing the total number of positive results—including those that were incorrectly identified—by the total number of positively recognised results. This means that it penalizes accuracy or recall extremes. Depending on which class is deemed positive, this metric is not symmetric across classes. Since it is TP-proportional, the F1 score would be high in this situation. The information and relative class distribution in this instance are both unchanged. In order to penalize excessive levels of

either, the F1-score is calculated as the harmonic mean (or weighted average) of accuracy and recall. Therefore, the applicability of this statistic depends on the class you're referring to. The F1 score would be high if the classifier were biased in favour of the large positive class. This score ranges from 0 to 1, where 1 denotes the highest accuracy and recall and 0 denotes no precision.

It's important to emphasize and examine important points (before the presentation of results). Here's what to do: When scientists combine two computer algorithms inspired by nature (like Alg1 and Alg2) to create a hybrid algorithm, they typically identify Alg1's shortcomings and address them using Alg2's advantages.

This hybrid approach first optimizes the set of features before classifying data using one or more machine learning classifiers using the optimized subset of features.

This differs from our approach, though. In our case, we first gave Alg1 the complete set of extracted features in the hope that the best accuracy would be obtained with a smaller set of features. The absolute maximum accuracy is thought to be Alg1. In order to further reduce the collection of features, Alg2 is then given the subset of features that Alg1 has returned. The Alg1 rejected features won't be taken into account by Alg2. Two layers were therefore used: the first layer used an optimized subset of features to find the maximum accuracy that could be achieved, and the second layer further reduced the subset of features. As a result, rather than combining two algorithms, we do so by examining and enhancing each method's distinctive characteristics (Figs. 4 and 5). While significantly reducing the feature subset provided by the first technique, the second method frequently results in a small accuracy reduction. We did find, however, that in some situations, when the second method is used, the accuracy raises more when the feature subset is shrunk. This is because some attributes (features) are detrimental for accuracy, and by removing them in the second round (i.e., Alg2), the feature set is further reduced while accuracy is improved. In addition, to find the best answer in our case, we considered a number of options. Suppose we are considering three machine learning algorithms (ML1, ML2, and ML3), then we have investigated all the options in our two-layer method, namely Alg1-ML1-Alg2-ML1, Alg1-ML1-Alg2-ML2, Alg1-ML1-Alg2-ML3, Alg1-ML2-Alg2-ML1, Alg1-ML2-Alg2-ML2, Alg1-ML2-Alg2-ML3, Alg1-ML3-Alg2-ML1, Alg1-ML3-Alg2-ML2, Alg3-ML1-Alg2-ML3. One could argue that the problem at hand can be formulated as a multi-objective problem with the ultimate goal of achieving maximum accuracy with the fewest possible features. A pareto front can be constructed from which the solution(s) can be extracted based on the requirements and trade-offs. However, we have viewed this problem from a different angle, where the first layer (Alg₁ objective)'s is to achieve maximum accuracy with features subset, and the second layer (Alg₂ objective)'s is to further reduce the feature subset, so we are offering another solution in which the features are further reduced with a trade-off in accuracy.

$$\text{Sensitivity} = \frac{\text{True positives}}{\text{True Positives} + \text{False Negatives}} \quad (7)$$

$$\text{Specificity} = \frac{\text{True negative}}{\text{True Negatives} + \text{False positives}} \quad (8)$$

$$\text{Precision} = \frac{\text{True positives}}{\text{True Positives} + \text{False Positives}} \quad (9)$$

$$Accuracy = \frac{\text{True positives} + \text{True Negatives}}{\text{True Positives} + \text{False Positives} + \text{False Negatives} + \text{True Negatives}} \quad (10)$$

$$F1 - Score = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (11)$$

Tables 7, 8, and 10 show how ABC, PSO, and BCS performed on REFUGE, ORIGA, and their combination. Three machine learning classifiers were used to generate three solutions for each dataset, and each table displays the results from all possible combinations. Similar to Table 8, Table 9 has been created, and it lists 27 potential solutions (nine solutions per dataset) for a two-layered approach. ABC follows PSO as the first algorithm that is used, with the output (features) of PSO serving as ABC's input. All potential solutions are generated for each subject dataset because there are three machine learning classifiers chosen for performance evaluation, making a maximum possible combination of nine (3*3). Table 12 was meticulously put together; it can be seen as a summary of the entire experiment, showing the two best results—one in terms of accuracy and the other in terms of feature count per dataset. We've also put forth trade-off solutions for two datasets: REFUGE and REFUGE plus ORIGA.

Tables 9 and 11, which represent data that the researchers hardly ever present, are two additional creative compilations that we've made. Table 9 shows the values of various performance measuring parameters during the entire 10,000 iterations when the number of features was minimum in a particular solution out of all feasible solutions. It was created from the 2-layered implemented approach of PSO followed by ABC. Similar to Table 10, Table 11 is also created for the implemented 2-layered approach; all the crucial parameters are reported in the table in the case of choosing the solution with the fewest features out of all viable solutions during the 10,000 executed iterations. The values of the performance measurement metrics shown in this table at that particular moment and the minimum number of features that can be included in the solution make it a crucial table. A comparison of the variations in parameter values between solutions with the highest accuracy and those with the fewest features is also possible. Table 12 was meticulously put together; it can be seen as a summary of the entire experiment, showing the two top results—one in terms of accuracy and the other in terms of feature count per dataset. Additionally, we have suggested a trade-off solution for two datasets: REFUGE and the union of REFUGE and ORIGA.

The unique Table 9 we created also contains data that is uncommonly published by academicians. Table 9 is compiled from the first 2-layered implemented approach, which included PSO followed by ABC, and it shows the critical data of the second implemented algorithm (ABC), showing the values of various performance measuring parameters during the times when the number of features in a particular solution was the smallest out of all feasible solutions over the course of the entire 10,000 iterations. This table is important because it shows the current values of the performance measurement metrics as well as the bare minimum number of features that can be included in the solution. Additionally, we can assess how much the most accurate solution and the solution with the fewest features differ in terms of parameter values. The two top options for each dataset in terms of accuracy and feature count are shown in the comprehensive compilation of Table 12, which can be viewed as a summary of the entire experiment. Additionally, we have offered a trade-off method for the REFUGE dataset and its combination with ORIGA.

Table 7 Performance of ABC on all selected datasets

S. No	Algorithm	DataSet	ML Classifier	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score
1	ABC	REFUGE	RF	32	93.7262	0.9434	0.9016	0.9325	0.9682
2	ABC	REFUGE	SVM	22	93.8117	0.9363	0.7933	0.9878	0.9614
3	ABC	REFUGE	ENSEM	27	94.2344	0.9361	0.9583	0.9981	0.9661
4	ABC	ORIGA	RF	29	93.0634	0.8831	0.9053	0.9666	0.9696
5	ABC	ORIGA	SVM	18	93.5103	0.9667	0.9090	0.9687	0.9677
6	ABC	ORIGA	ENSEM	36	93.9810	0.9808	0.8971	0.9625	0.9716
7	ABC	REFUGE+ORIGA	RF	24	93.2065	0.9667	0.9081	0.9687	0.9381
8	ABC	REFUGE+ORIGA	SVM	13	93.3268	0.9198	0.8468	0.9748	0.9465
9	ABC	REFUGE+ORIGA	ENSEM	26	92.9386	0.8840	0.8641	0.9716	0.9257

Table 8 Performance of PSO on all selected datasets

S. No	Algo	DataSet	ML Classifier	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score
1	PSO	REFUGE	RF	21	92.8679	0.9363	0.8833	0.9878	0.9599
2	PSO	REFUGE	SVM	24	93.7097	0.9361	0.9481	0.9981	0.9650
3	PSO	REFUGE	ENSEM	21	94.1275	0.9434	0.9016	0.9944	0.9673
4	PSO	ORIGA	RF	29	95.2012	0.9667	0.9051	0.9687	0.9660
5	PSO	ORIGA	SVM	19	95.8204	0.9808	0.8911	0.9625	0.9701
6	PSO	ORIGA	ENSEM	32	95.5108	0.9727	0.9053	0.9666	0.9696
7	PSO	REFUGE+ORIGA	RF	24	90.7960	0.9198	0.8968	0.9748	0.9465
8	PSO	REFUGE+ORIGA	SVM	15	86.8335	0.8840	0.8641	0.9716	0.9246
9	PSO	REFUGE+ORIGA	ENSEM	28	86.9423	0.8981	0.8190	0.9536	0.9240

Table 9 Performance of 2 layered PSO-ABC based approach on all selected datasets

S. No	Data Set	PSO Output ACC. (FEA.)	PSO-ML Class	ABC-ML Class	No. of Features	Accuracy	Sensit	Specif	Prec	F-1
1	REFUGE	92.8679 (21)	RF	RF	15	93.4329	0.9312	0.8561	0.9911	0.9599
2	REFUGE	92.8679 (21)	RF	SVM	13	91.5674	0.9289	0.9299	0.9921	0.9599
3	REFUGE	92.8679 (21)	RF	ENS	17	92.9213	0.9212	0.8701	0.9909	0.9612
4	REFUGE	93.7097 (24)	SVM	RF	15	94.2131	0.9301	0.831	0.9943	0.9699
5	REFUGE	93.7097 (24)	SVM	SVM	11	93.5399	0.9399	0.941	0.9898	0.9987
6	REFUGE	93.7097 (24)	SVM	ENS	17	93.3001	0.9399	0.8712	0.9901	0.9687
7	REFUGE	94.1275 (21)	ENS	RF	18	92.6225	0.9121	0.7909	0.9799	0.9498
8	REFUGE	94.1275 (21)	ENS	SVM	8	92.2089	0.9112	0.8419	0.9933	0.961
9	REFUGE	94.1275(21)	ENS	ENS	11	93.8787	0.9399	0.8801	0.9944	0.9774
10	ORIGA	95.2012 (29)	RF	RF	19	96.211	0.94675	0.9034	0.9587	0.9788
11	ORIGA	95.2012 (29)	RF	SVM	21	95.3272	0.9677	0.8901	0.9499	0.9593
12	ORIGA	95.2012 (29)	RF	ENS	16	95.4001	0.9687	0.9011	0.9521	0.9588
13	ORIGA	95.8204 (19)	SVM	RF	9	94.3134	0.9601	0.9101	0.9534	0.9588
14	ORIGA	95.8204 (19)	SVM	SVM	11	95.1122	0.9801	0.8902	0.9621	0.9609
15	ORIGA	95.8204 (19)	SVM	ENS	12	95.5496	0.9612	0.8943	0.9609	0.9666
16	ORIGA	95.5108 (32)	ENS	RF	19	93.5122	0.9654	0.8921	0.9629	0.9621
17	ORIGA	95.5108 (32)	ENS	SVM	19	95.0001	0.9801	0.9099	0.9621	0.9709
18	ORIGA	95.5108 (32)	ENS	ENS	21	94.9988	0.9721	0.9098	0.9621	0.97
19	REFUGE+ORIGA	90.796 (24)	RF	RF	12	82.0005	0.9099	0.8076	0.9778	0.9446
20	REFUGE+ORIGA	90.796 (24)	RF	SVM	13	86.4643	0.8712	0.7927	0.9761	0.9249
21	REFUGE+ORIGA	90.796 (24)	RF	ENS	9	87.4536	0.8911	0.7498	0.9699	0.9321
22	REFUGE+ORIGA	86.8335 (15)	SVM	RF	10	80.3456	0.9199	0.7943	0.9699	0.9532
23	REFUGE+ORIGA	86.8335 (15)	SVM	SVM	10	87.1909	0.8673	0.7777	0.9712	0.9211
24	REFUGE+ORIGA	86.8335 (15)	SVM	ENS	11	87.2002	0.8602	0.8205	0.9267	0.9076
25	REFUGE+ORIGA	86.9423 (28)	ENS	RF	15	89.1122	0.9212	0.7849	0.9709	0.9498
26	REFUGE+ORIGA	86.9423 (28)	ENS	SVM	17	85.4312	0.8687	0.7401	0.9801	0.9101
27	REFUGE+ORIGA	86.9423 (28)	ENS	ENS	11	86.0801	0.8799	0.7999	0.9587	0.9123

Table 10 Performance of BCS based approach on all selected datasets

S. No	Algorithm	DataSet	ML Classifier	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score
1	BCS	REFUGE	RF	42	94.5606	0.9424	0.9076	0.9506	0.9606
2	BCS	REFUGE	SVM	39	96.2343	0.9136	0.9181	0.9562	0.9531
3	BCS	REFUGE	ENSEM	39	95.3974	0.9333	0.9269	0.9516	0.9685
4	BCS	ORIGA	RF	47	98.4615	0.9409	0.8838	0.9625	0.9515
5	BCS	ORIGA	SVM	41	96.9230	0.9543	0.8780	0.9583	0.9563
6	BCS	ORIGA	ENSEM	42	95.3846	0.9524	0.8827	0.9604	0.9564
7	BCS	REFUGE+ORIGA	RF	36	94.5652	0.9108	0.8801	0.9671	0.9677
8	BCS	REFUGE+ORIGA	SVM	40	91.8478	0.8639	0.8931	0.9285	0.8950
9	BCS	REFUGE+ORIGA	ENSEM	38	91.5760	0.8748	0.8875	0.9362	0.9045

Table 11 Performance of 2 layered PSO-ABC based approach, with minimum number of features, on all selected datasets

S. No	Data set	PSO-ML Class	ABC-ML Class	Minimum No. of Features	Accuracy	Sen	Spec	Prec	F-1
1	REFUGE	RF	RF	4	91.6107	0.9286	0.7666	0.9822	0.9547
2	REFUGE	RF	SVM	5	92.2813	0.9261	0.8292	0.9934	0.9586
3	REFUGE	RF	ENS	4	92.1979	0.9268	0.7954	0.9916	0.9581
4	REFUGE	SVM	RF	4	90.8347	0.9318	0.7657	0.9692	0.9502
5	REFUGE	SVM	SVM	4	89.9093	0.9051	0.8002	0.9150	0.8986
6	REFUGE	SVM	ENS	4	92.9530	0.9356	0.8070	0.9897	0.9619
7	REFUGE	ENS	RF	4	91.4431	0.9319	0.7885	0.9822	0.9564
8	REFUGE	ENS	SVM	4	84.4940	0.8449	0.7965	0.9234	0.9159
9	REFUGE	ENS	ENS	7	92.9527	0.9349	0.8181	0.9906	0.9619
10	ORIGA	RF	RF	4	93.5005	0.9425	0.8679	0.9562	0.9493
11	ORIGA	RF	SVM	4	94.5836	0.9664	0.8875	0.9604	0.9634
12	ORIGA	RF	ENS	12	94.4300	0.9565	0.8895	0.9625	0.9595
13	ORIGA	SVM	RF	6	93.6570	0.9483	0.8703	0.9562	0.9522
14	ORIGA	SVM	SVM	11	95.8232	0.9726	0.8947	0.9625	0.9675
15	ORIGA	SVM	ENS	6	94.5844	0.9564	0.8841	0.9604	0.9584
16	ORIGA	ENS	RF	6	93.4933	0.9506	0.8875	0.9625	0.9565
17	ORIGA	ENS	SVM	8	95.6609	0.9746	0.8901	0.9604	0.9674
18	ORIGA	ENS	ENS	6	94.7394	0.9584	0.8848	0.9604	0.9594
19	REFUGE+ORIGA	RF	RF	3	79.7079	0.8425	0.7767	0.9201	0.8796
20	REFUGE+ORIGA	RF	SVM	4	84.4940	0.8449	0.7802	0.9082	0.9159
21	REFUGE+ORIGA	RF	ENS	4	84.5483	0.8454	0.7901	0.9260	0.9162
22	REFUGE+ORIGA	SVM	RF	3	79.3270	0.8368	0.7275	0.9317	0.8817
23	REFUGE+ORIGA	SVM	SVM	4	84.4940	0.8449	0.8301	0.8910	0.9159
24	REFUGE+ORIGA	SVM	ENS	3	82.0902	0.8106	0.7909	0.9012	0.9197
25	REFUGE+ORIGA	ENS	RF	3	77.9138	0.8371	0.7629	0.9137	0.8737
26	REFUGE+ORIGA	ENS	SVM	4	84.4940	0.8449	0.8034	0.8901	0.9159
27	REFUGE+ORIGA	ENS	ENS	3	84.5418	0.8454	0.8099	0.8197	0.9162

Table 12 Summary sheet of the performance

The best performing algorithm in terms of accuracy in case of REFUGE dataset									
S. No	Algo	DataSet	ML Classifier	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score
1	BCS	REFUGE	SVM	39	96.2343	0.9136	0.8181	0.9962	0.9531
The best performing algorithm in terms of features count in case of REFUGE dataset									
S. No	Algorithm + ML Classifier	DataSet	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score	
2	PSO-SVM-ABC-ENS	REFUGE	4	92.9530	0.9356	0.8070	0.9897	0.9619	
Alternate Solution-The trade-off between accuracy and features count in case of REFUGE dataset									
S. No	Algorithm + ML Classifier	DataSet	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score	
3	PSO-SVM-ABC-RF	REFUGE	15	94.2131	0.9301	0.8310	0.9943	0.9699	
The best performing algorithm in terms of accuracy in case of ORIGA dataset									
S. No	Algorithm	DataSet	ML Classifier	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score
4	BCS	ORIGA	RF	47	98.4615	0.9409	0.8838	0.9625	0.9515
The best performing algorithm in terms of features count in case of ORIGA dataset									
S. No	Algorithm + ML Classifier	DataSet	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score	
5	PSO-RF-ABC-SVM	ORIGA	4	94.5836	0.9664	0.8875	0.9604	0.9634	
6	PSO-SVM-ABC-SVM	ORIGA	11	95.8232	0.9726	0.8947	0.9625	0.9675	
Alternate Solution-The trade-off between accuracy and features count in case of REFUGE + ORIGA dataset									
S. No	Algorithm + ML Classifier	DataSet	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score	
7	PSO-RF-ABC-RF	ORIGA	19	96.2110	0.94675	0.9034	0.9587	0.9788	
The best performing algorithm in terms of accuracy in case of REFUGE + ORIGA dataset									
S. No	Algorithm	DataSet	ML Classifier	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score
55	BCS	REFUGE+ORIGA	RF	36	94.5652	0.9108	0.7801	0.9671	0.9381
The best performing algorithm in terms of features in case of REFUGE+ORIGA dataset									
S. No	Algorithm + ML Classifier	DataSet	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score	
9	PSO-ENS-ABC-ENS	REFUGE + ORIGA	3	84.5418	0.8454	0.8099	0.8197	0.9162	
Alternate Solution-The trade-off between accuracy and features count in case of REFUGE + ORIGA dataset									
S. No	Algorithm + ML Classifier	DataSet	No. of Features	Accuracy	Sensitivity	Specificity	Precision	F-1 Score	
10	ABC-SVM	REFUGE+ORIGA	13	93.3268	0.9198	0.7968	0.9748	0.9465	
11	PSO-RF	REFUGE+ORIGA	24	90.7960	0.9198	0.8968	0.9748	0.9465	
12	PSO-RF-ABC-RF	REFUGE+ORIGA	12	91.0005	0.9099	0.8076	0.9778	0.9446	

The choice of features (feature selection (FS)) will be crucial in the creation of a computer-based glaucoma detection system. FS is the problem of finding the smallest subset of features that accurately represents the original feature set. So, reducing the number of features in a feature set is also known as feature selection (FS). By eliminating superfluous or pointless features, these techniques reduce the dimensionality of the feature space. Misclassification could be reduced with the aid of computer-based diagnostic techniques. We employ feature selection techniques to produce a feature set with high discriminating ability. As opposed to glaucoma, which may have minor, highly correlated, or redundant features, such systems need vast datasets with a

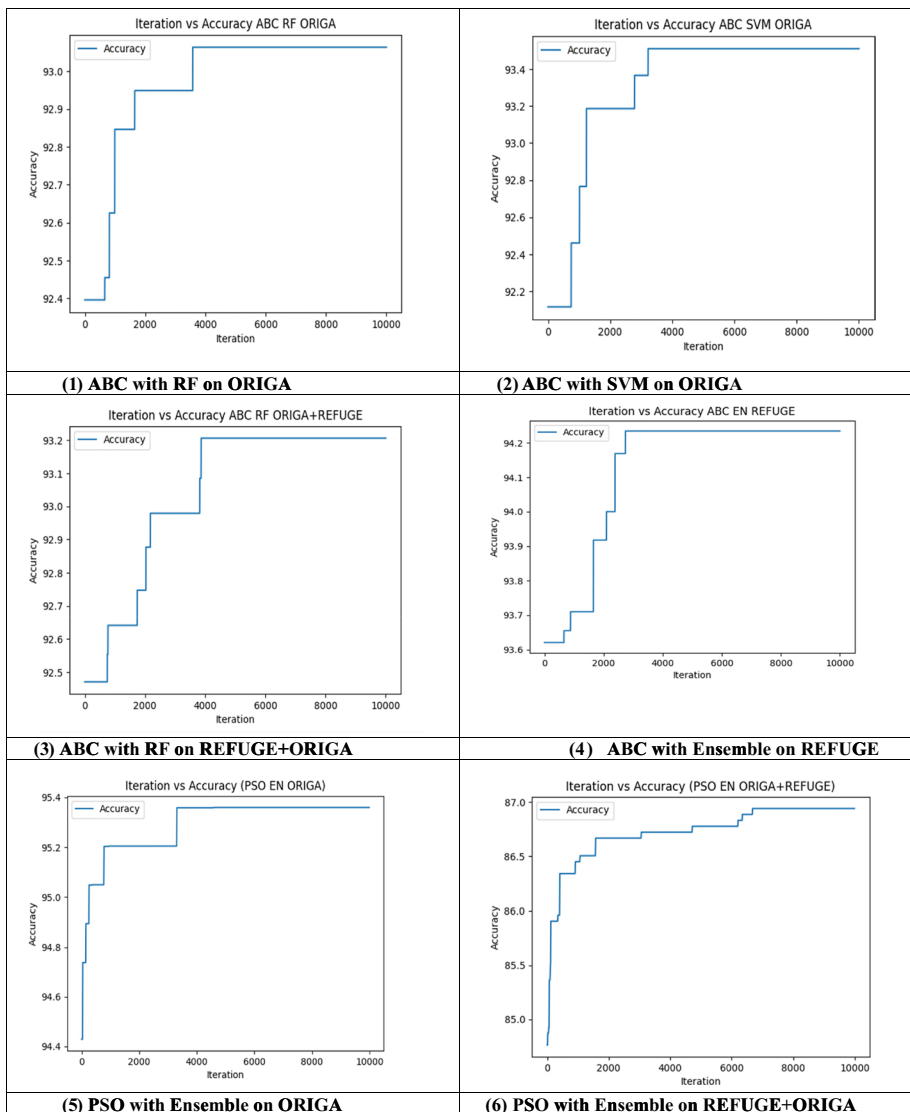


Fig. 9 Accuracy vs iteration graphs of various algorithms on different datasets

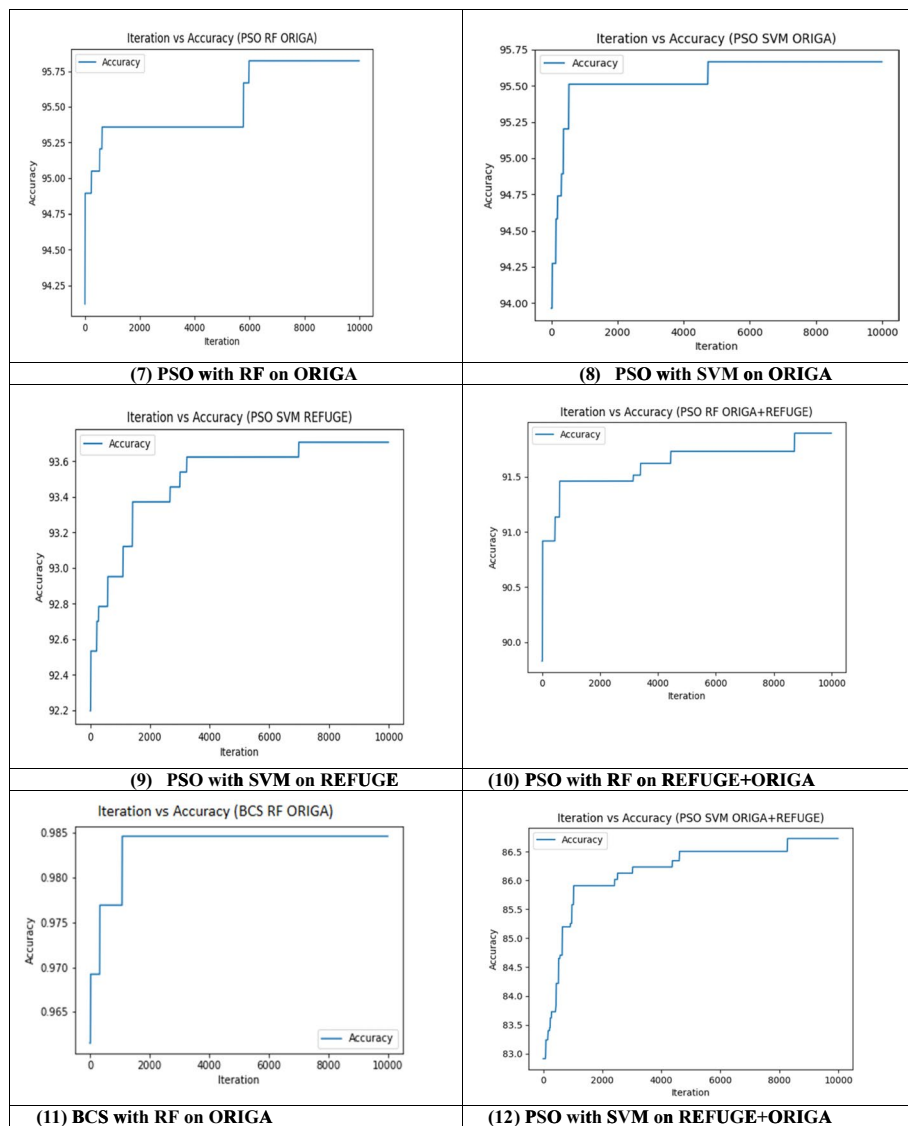


Fig. 9 (continued)

variety of cases and characteristics to train. To reduce the size of the feature space and find appropriate features with high discriminative power, feature selection techniques are therefore used before classification. For machine learning to effectively remove redundant features from enormous amounts of real-world data, FS is now required. Machine learning algorithms' effectiveness and accuracy may be hampered by unnecessary and pointless features. By eliminating unnecessary and stale features and focusing on those that would improve the system's performance, they try to determine the optimal subset. Researchers in operational and artificial intelligence have been

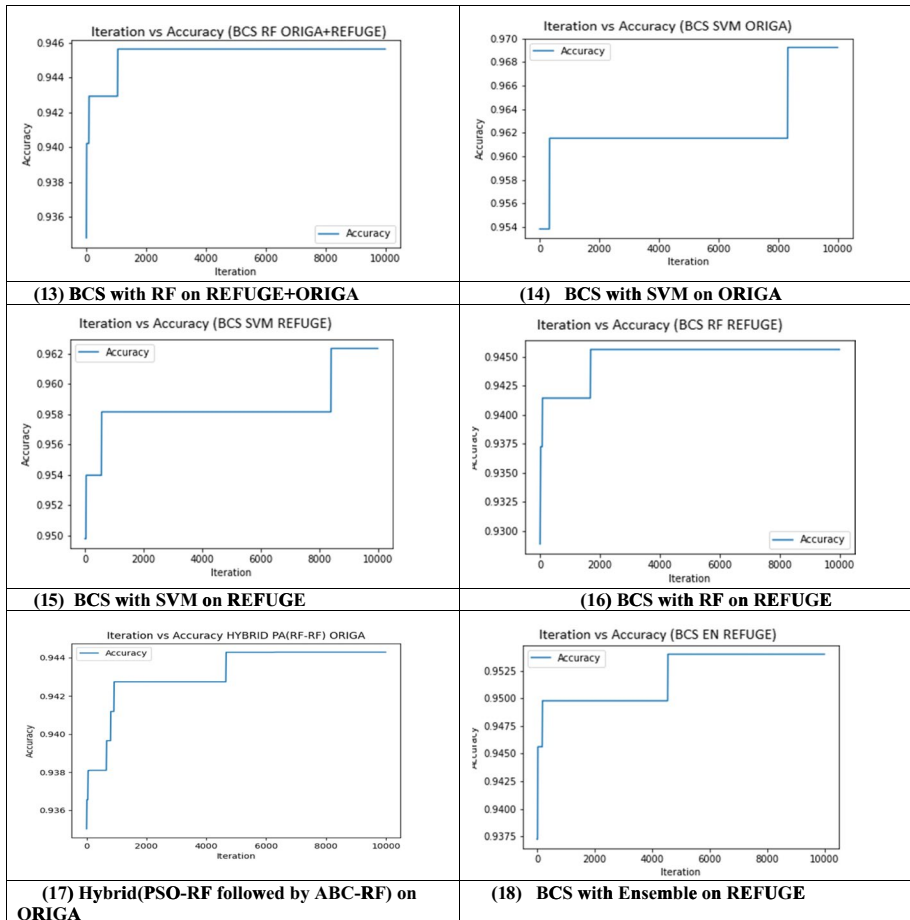


Fig. 9 (continued)

interested in this field since the 1970s due to its applications. In addition to speeding up classification while reducing feature count, FS methods also improve calculation efficiency. If we examine these tables, we can see that the BCS algorithm produces the best results in terms of accuracy, and that the feature count that BCS chooses to produce the best accuracy is also the highest in comparison to the other algorithms. The two-layered approach made up of PSO-ABC performs the best in terms of feature count. With the help of our methodology, we are able to achieve astounding results, such as 96.23% accuracy (with 39 features) for the REFUGE dataset, 98.46% accuracy (with 47 features) for the ORIGA dataset, and 94.56% accuracy (with 36 features) for a combination of the REFUGE and ORIGA datasets. If we look at the results from the point of view of feature count, we see that our method is very fruitful, as we were able to generate 92.2089% accuracy using 8 features in the case of the REFUGE dataset, up to 94.3134% accuracy with just 9 features in the case of the ORIGA dataset, and finally, 93.32% accuracy with 13 features (20% of the initial count of 65 features)

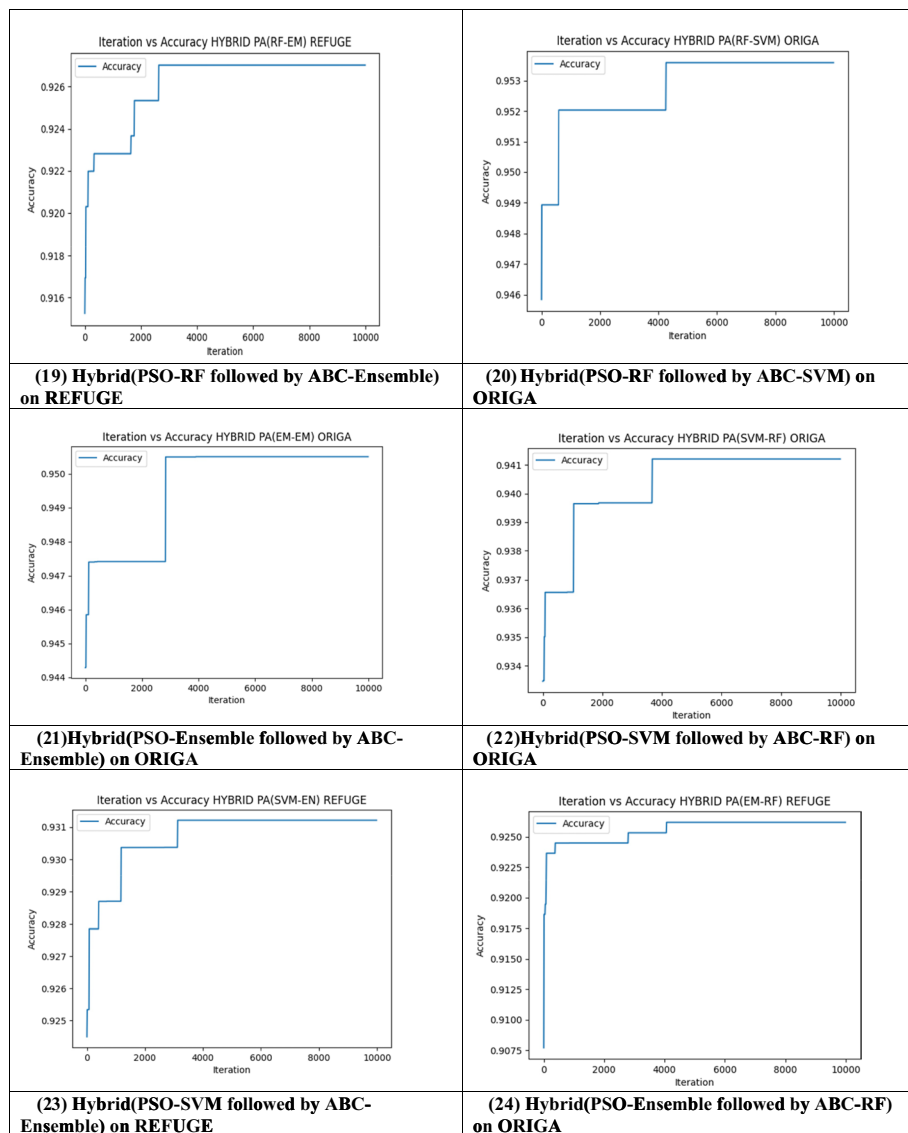


Fig. 9 (continued)

only in case of combination of both datasets. The ABC and PSO algorithms produced the best sensitivity of 0.9434 in REFUGE, while BCS lagged these two by a very small amount. The combined sensitivity produced by ABC and PSO in ORIGA was at its highest, 0.9808, and the best sensitivity was produced by the ABC algorithm at 0.9667 in this combination of two datasets. The ABC algorithm produced the best specificity in REFUGE, which was 0.9583. The ABC algorithm generated the highest specificity possible for ORIGA, 0.9090; similarly, when these two datasets were combined, the ABC algorithm produced the best specificity possible, 0.9081. BCS's

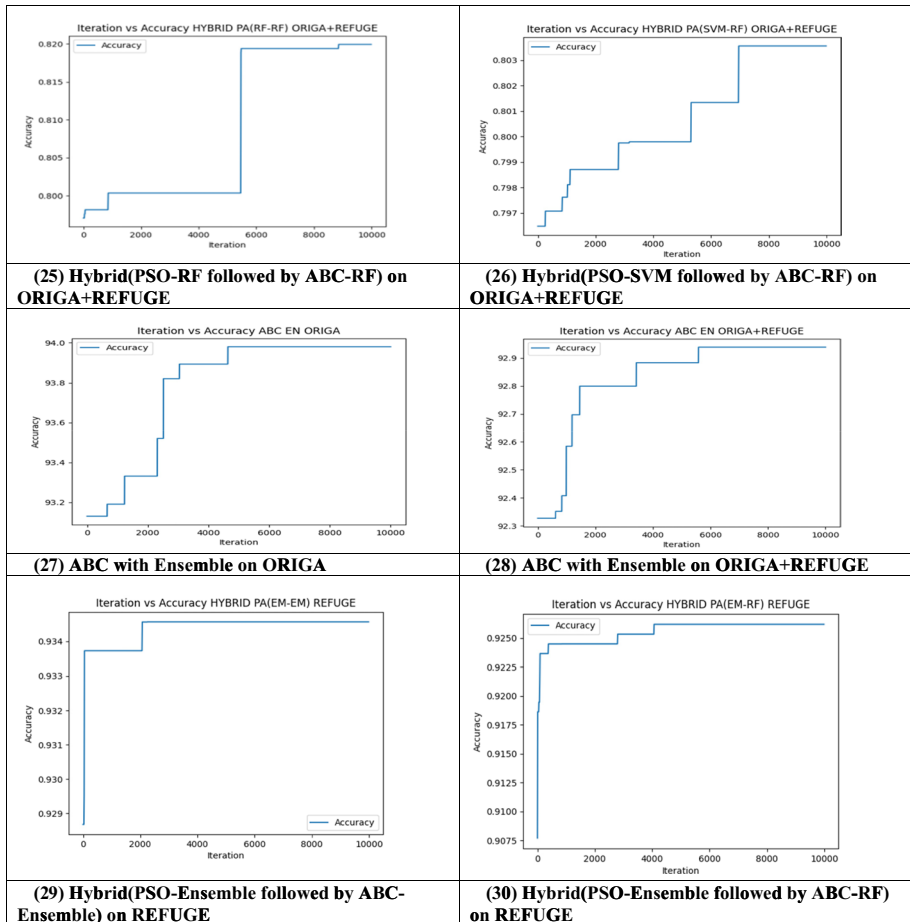


Fig. 9 (continued)

algorithm produced the best F1-score in REFUGE, which was 0.9685. The highest F1-score produced by the ABC algorithm for ORIGA was 0.9701; similarly, the BCS algorithm produced the highest F1-score from the combination of these two datasets, which was 0.9677.

Out of the possible 54 cases (9+9+9+27) of accuracy vs iteration graphs, we hereby presented 30 graphs(greater than 50% of the total graphs, due to space constraints) representing 30 cases, Fig. 9, depicting how accuracy has been improved iteration wise.

Out of the possible 54 possible cases of number_of_features vs iteration graphs, we have hereby presented 12 graphs(due to space constraints) representing 12 cases depicting features count and accuracy at every 1000 iteration (Fig. 10).

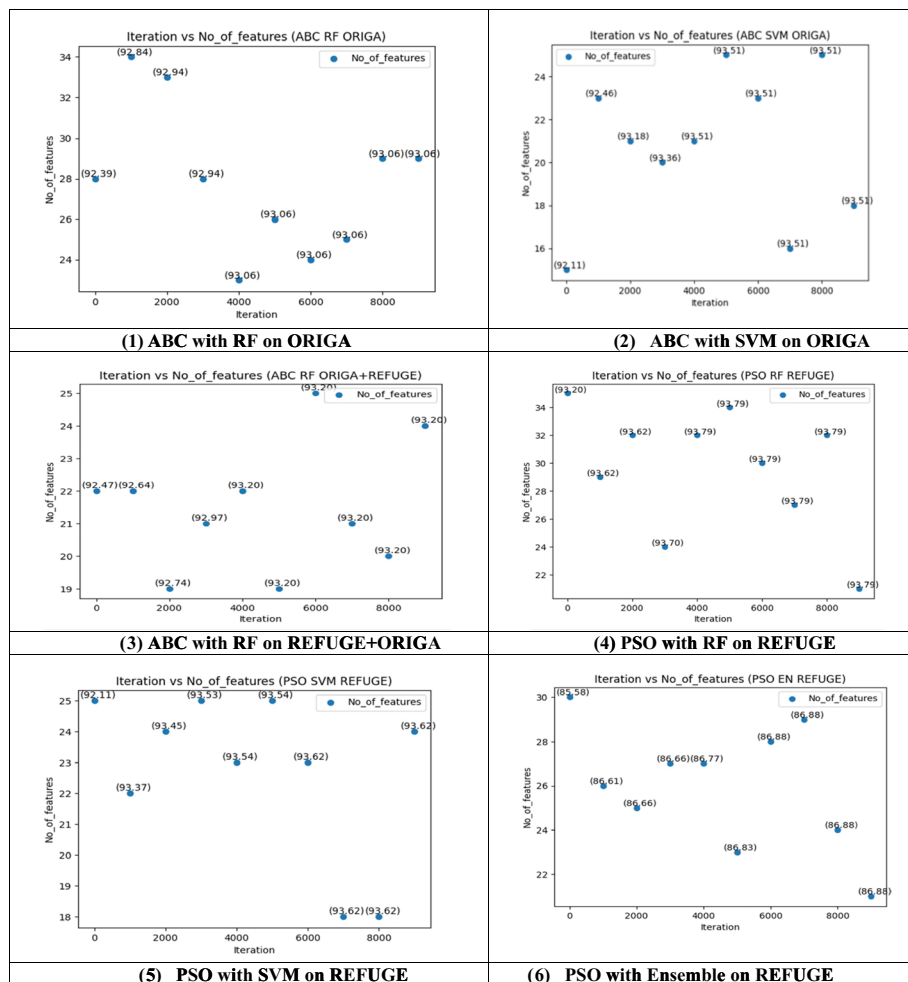


Fig. 10 Number_of_features vs iteration graphs

5 Comparison with prior published literature

The suggested approach is compared to the most popular current glaucoma diagnostic techniques in Table 13. This table provides irrefutable evidence that the proposed novel approach is robust and efficient in categorizing fundus images with an accuracy of up to 98.4615%. As with other metrics, the sensitivity and specificity of the performance are both high. One can see that our performance is remarkable when compared to the other 20 approaches (shown in Table 13 published in or after 2017). Our method (which is based on soft computing and machine learning) has shown to be more effective than deep learning methods in many real-world scenarios. However, there may be a few instances when the datasets on which the other methods were evaluated are different. As we have already established, we have not restricted ourselves to only two datasets in our research; rather, we have experimented with combination of many datasets

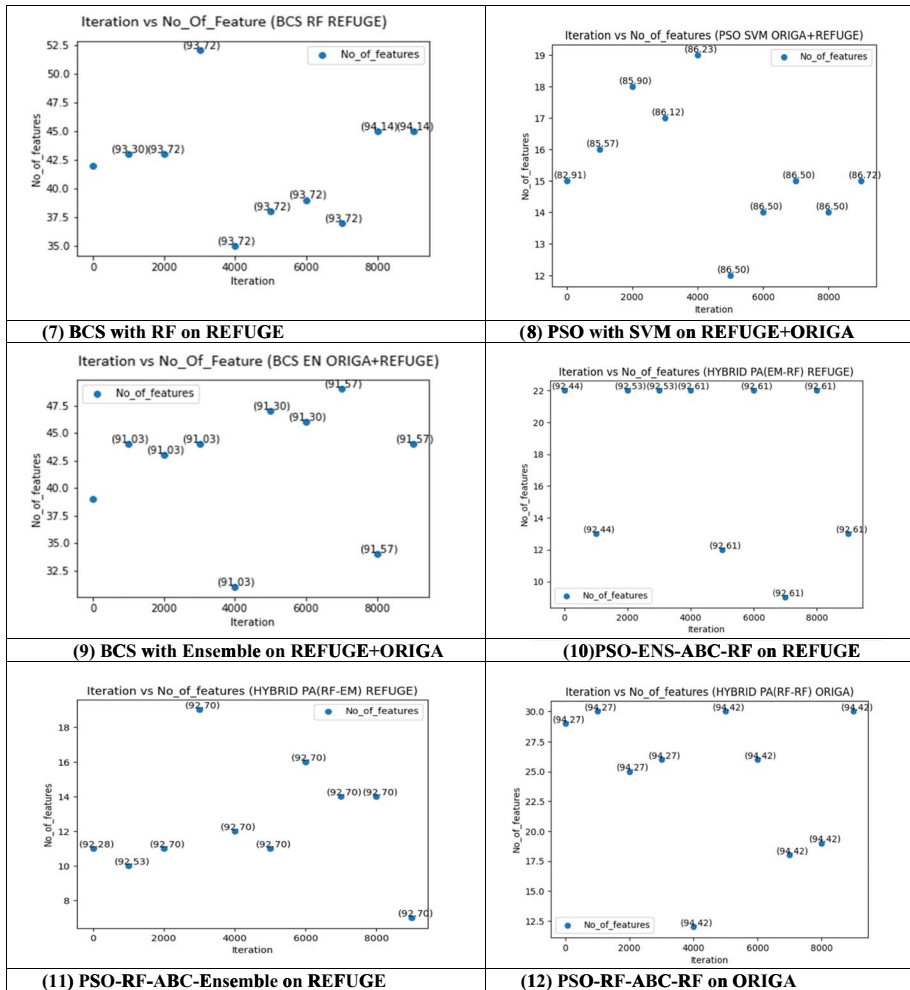


Fig. 10 (continued)

to learn more about how well our proposed technique performs in cross-validation. This indicates the robustness and generalizability of our suggested method.

6 Conclusion

The goal of this study is to find a way to classify images based on soft computing and machine learning that uses only a small subset of the initial 65 features needed to find glaucoma in retinal images. For this research, we took information from not one but two benchmark datasets. We used three well proven soft computing techniques and one soft computing strategy with two layers to improve the way features work. As far as we are aware, our 65 extracted traits have not been extensively utilized to classify glaucoma using any of these four methodologies. The fact that we ran our trials on each of these

Table 13 A comparative study of the proposed method's performance in comparison to other work for glaucoma diagnosis

Study	Methods/Features	Classification method	No. of images	ACC	SEN	SPE
[90] 2021	Convolution neural network and deep-belief network (DBN)	DL based approach	Private Images collected from hospitals	98.00	0.83	0.97
[53] 2020	Self composed CNN Glaucoma network (G-net)	DL based approach	DRISHTI-DS	95.03	0.7379	0.8623
[33] 2021	Self Crafted CNN	DL based approach	ORIGA	78.32	0.5806	0.9244
[122] 2021	Custom UNET++ model	DL based approach	650 images of ORIGA	90.00	0.87	0.81
[34] 2021	BEMD algorithm; VGG-19 for extraction of features from decomposed BEMD components, PCA applied for features selection	CNN's features + SVM	ORIGA	96.67	0.9665	0.9671
[94] 2018	18 layered self composed CNN network for glaucoma detection	DL based approach	Private Data set	98.13	0.9800	0.9830
[95] 2018	Optic disk localization and non-parametric GIST descriptor	SVM and 19 features (Radon + MCT + GIST + LSDA)	1000 Private and Public images	97.00	0.8750	0.9843
[121] 2020	Extracted Structural and Non-structural features. Followed by many ML classifiers for classification	SVM performs the best	RIM-1 and DRISHTI-DS	97.22	0.95	0.95
[26] 2017	Local Binary Pattern (LBP) to represent the optic disk region, and geostatistical functions to describe texture patterns	Micro-Genetic Algorithm and SVM	455 images of RIM-1	91.20	0.9500	0.8820

Table 13 (continued)

Study	Methods/Features	Classification method	No. of images	ACC	SEN	SPE
[100] 2019	Hybrid classification approach consisting of Naïve Bayes binary classifier in the first stage followed by simplified multiclass support vector machine (Sim-MSVM)	Feature extractions and classification using inbuilt and customized ML classifiers	100 images of RIM-1	Upto 96.00	Upto 0.940	Upto 0.980
[69] 2017	Variational Mode Decomposition Method	Feature normalization, selection, and ranking; LS-SVM	488 private images	95.19	0.9362	0.9671
[42] 2020	UNet++ for segmentation and clinical features extraction. Increasing field of view (IFOV) for extraction of image based features. At last features optimization	Gradient boosting decision tree (GBDT)	650 images of ORIGA	84.30	0.894	0.793
[85] 2020	Optic disc/cup segmentation and glaucoma classification using DL	DL based approach	REFUGE	Not Reported	0.9752	Not Reported
[110] 2019	RCNN	Deep Learning (DL) based approach	ORIGA	Not Reported	0.7117	Not Reported
[41] 2018	RF+SMOTE	Machine Learning	ORIGA	76.90	0.799	0.738
[67] 2019	A semantic segmentation approach employing conditional GANs	DL based approach	650 images of ORIGA dataset and 400 images from REFUGE dataset	ORIGA 76.57 REFUGE 82.78	ORIGA 0.7273 REFUGE 0.7	ORIGA 0.8041 REFUGE 0.956
[37] 2019	DENet: Disc-aware ensemble network	DL based approach	ORIGA and Singapore Chinese Eye Study	84.29	0.8478	0.8380

Table 13 (continued)

Study	Methods/Features	Classification method	No. of images	ACC	SEN	SPE
[73] 2020	OD and OC Segmentation; classification using self crafted CNNs (MobileNetV2 for feature extraction)	DL based approach	2482 images from ORIGA and other datasets	88.00	0.83	0.90
[116] 2020	CNN (P1) pretrained CNNs for trans- fer learning; (P2) pretrained deep CNNs as the feature extractors; (P3) an ensemble of methods, P1 and P2	11 DL models, then SVM and finally their ensemble	2787 images from REFUGE, ACRIMA, ORIGA and other datasets	From 80.00 (ORIGA) to 95.59 (REFUGE)	Not Reported	Not Reported
[36] 2021	CNN	DL based approach	10,658 images from REFUGE, and other datasets	95.3	0.841	0.958
Proposed method	65 features belonging to various categories. Soft-computing approaches for optimization of features count	With machine learning approaches (SVM, RF and Ensemble)	2 Public Datasets (REFUGE and ORIGA) upto 1850 images	Upto 98.4615	Upto 0.9808	Upto 0.9583

datasets separately and in tandem is one of the intriguing parts of our experimental work. Pre-processing to remove outliers, an optic and an optic cup to locate CDR and DDLS, a feature extraction approach to extract the potential features, feature selection to create a subset of the most important features, and classification of retinal images as "normal" or "abnormal" based on training provided by features as input and labels as output are all included in the methodology. The input retinal images were processed to extract a total of 65 features(attributes), including structural and non-structural features including DDLS, CDR, GLRM, GLCM, HOS, FoS, HOC, and wavelets. These attributes (features) are then condensed using three widely used soft computing techniques (PSO, ABC, and BCS) and a proposed two-layered approach. These reduced features (feature selection) are then input to SVM, RF, and an ensemble of three classifiers for classification into two classes: normal and infected. By showing that a more focused collection of features might result in favourable results, the experimental results support our technique. Our efforts provide the scientific community with a more compact set of features in a shorter amount of time. For each situation, our efforts provide a variety of solutions; the anonymous end user may choose the one that best satisfies their needs for accuracy and feature count. Because of its better accuracy, the BCS algorithm outperforms competing techniques. This study can be made into software or a system that is accessible through the cloud or mobile devices as a contribution to human society, and it can also be used by ophthalmologists as a second opinion in the diagnosis of glaucoma for extensive clinical testing where there aren't enough qualified medical professionals to use the system or software. We want to make this research more useful so it can find more retinal diseases like diabetic retinopathy, maculopathy, retinal detachment, age-related macular degeneration, and so on. Using deep learning techniques, it will soon be possible to look at a huge number of images at once. In the future, we intend to build a convolutional neural network (CNN) model utilizing a multi-column, multi-scale CNN to detect the various phases of glaucoma using a vast database of altered photos as opposed to raw images. In the near future, we want to develop an automated glaucoma diagnosis approach utilizing a massive size database. Typical glaucoma classifications include images of the mild, moderate, and severe phases. In a later stage of evolution, we shall classify these stages.

Availability of data and material The data that support the findings of this study are available from the first author upon reasonable request.

Declarations

Conflicts of interest/Competing interests The authors declare no conflict of interest.

Informed consent None.

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